

LM Pre-training & Distillation

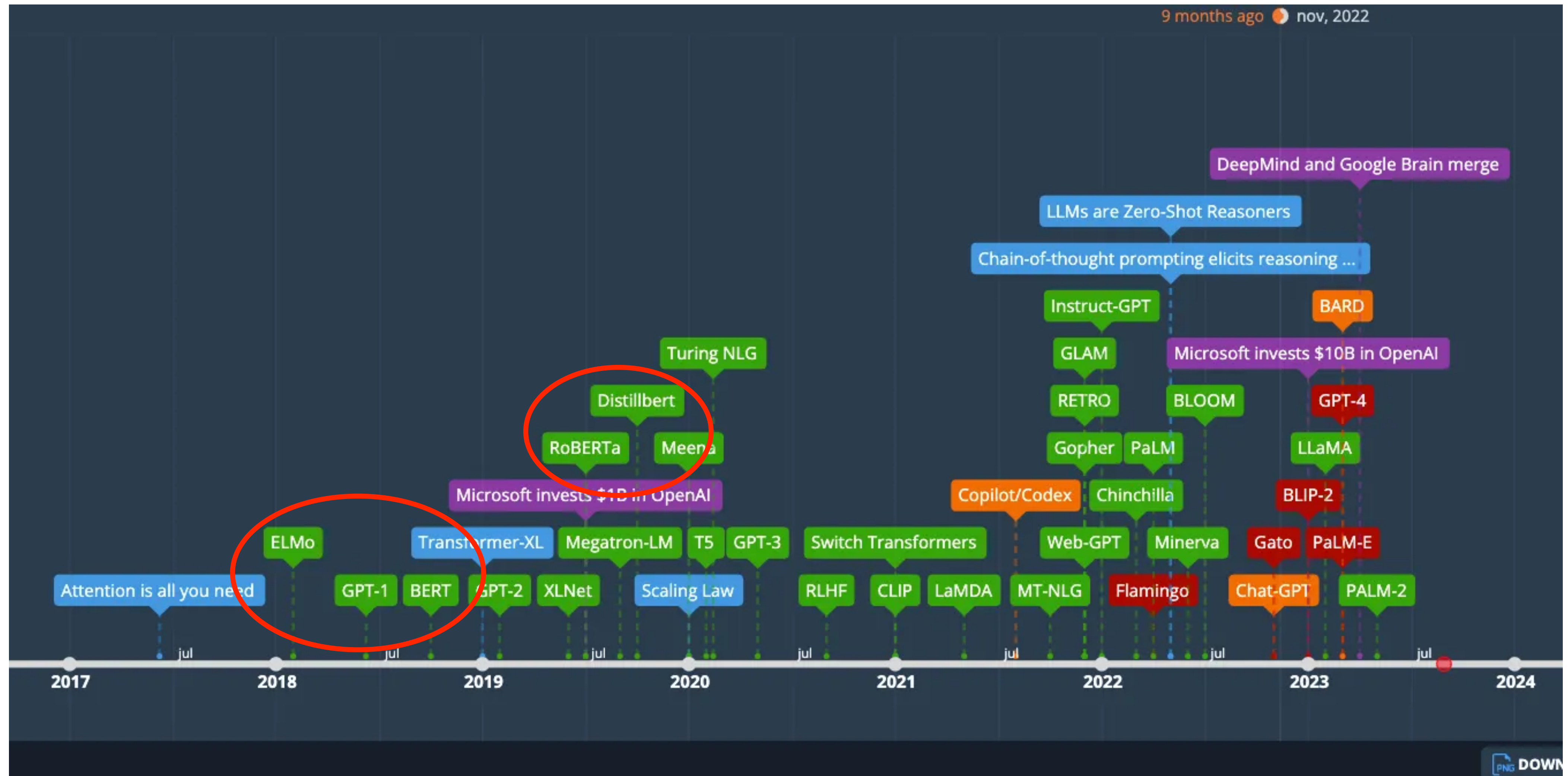
Wei Xu

(many slides from Greg Durrett)

This Lecture

- ▶ ELMo
- ▶ BERT, Results/Visualization, Extensions
- ▶ Knowledge Distillation

The beginning of LLMs



Readings

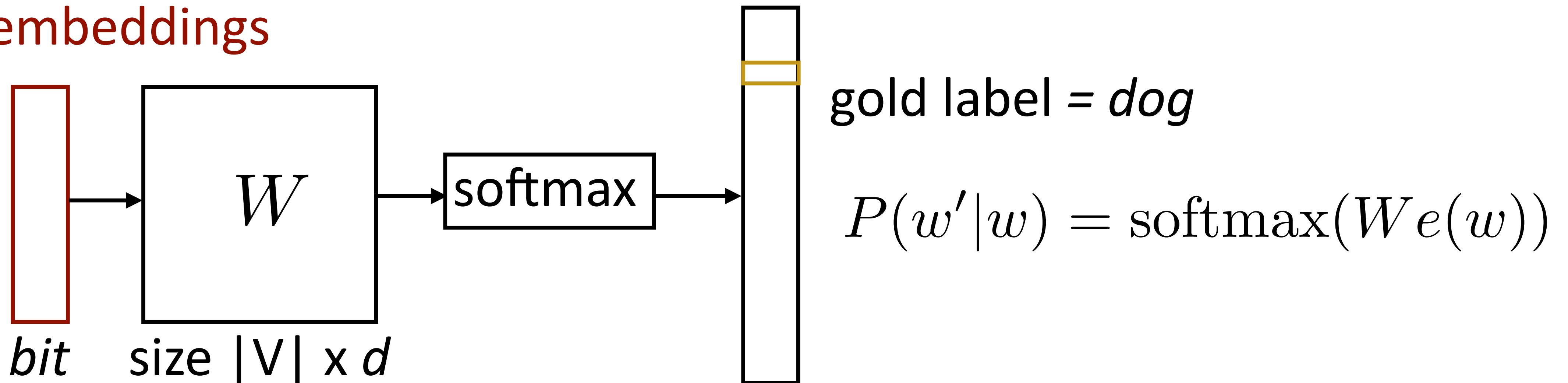
- ▶ Readings —
 - ▶ J+M 10
 - ▶ ELMo by Peters et al.
<https://aclanthology.org/N18-1202.pdf>
 - ▶ BERT by Devlin et al.
<https://aclanthology.org/N19-1423.pdf>

Recall: word2vec (Skip-Gram)

- Predict one word of context from word

d-dimensional
word embeddings

the dog bit the man



- Another training example: *bit* $\rightarrow$ *the*
- Parameters: $d \times |V|$ **vectors**, $|V| \times d$ output parameters (W) (also usable as vectors!)

ELMo

(released 2018/2, early version 2017/10)

ELMo

Deep contextualized word representations

Matthew E. Peters[†], Mark Neumann[†], Mohit Iyyer[†], Matt Gardner[†],
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Abstract

We introduce a new type of *deep contextualized* word representation that models both (1) complex characteristics of word use (e.g., syntax and semantics), and (2) how these uses vary across linguistic contexts (i.e., to model polysemy). Our word vectors are learned functions of the internal states of a deep bidirectional language model (biLM), which is pre-trained on a large text corpus. We show that these representations can be easily added to existing models and significantly improve the state of the art across six challenging NLP problems, including question answering, textual entailment and sentiment analysis. We also present an analysis showing that exposing the deep internals of the pre-trained network is crucial, allowing downstream models to mix different types of semi-supervision signals.

1 Introduction

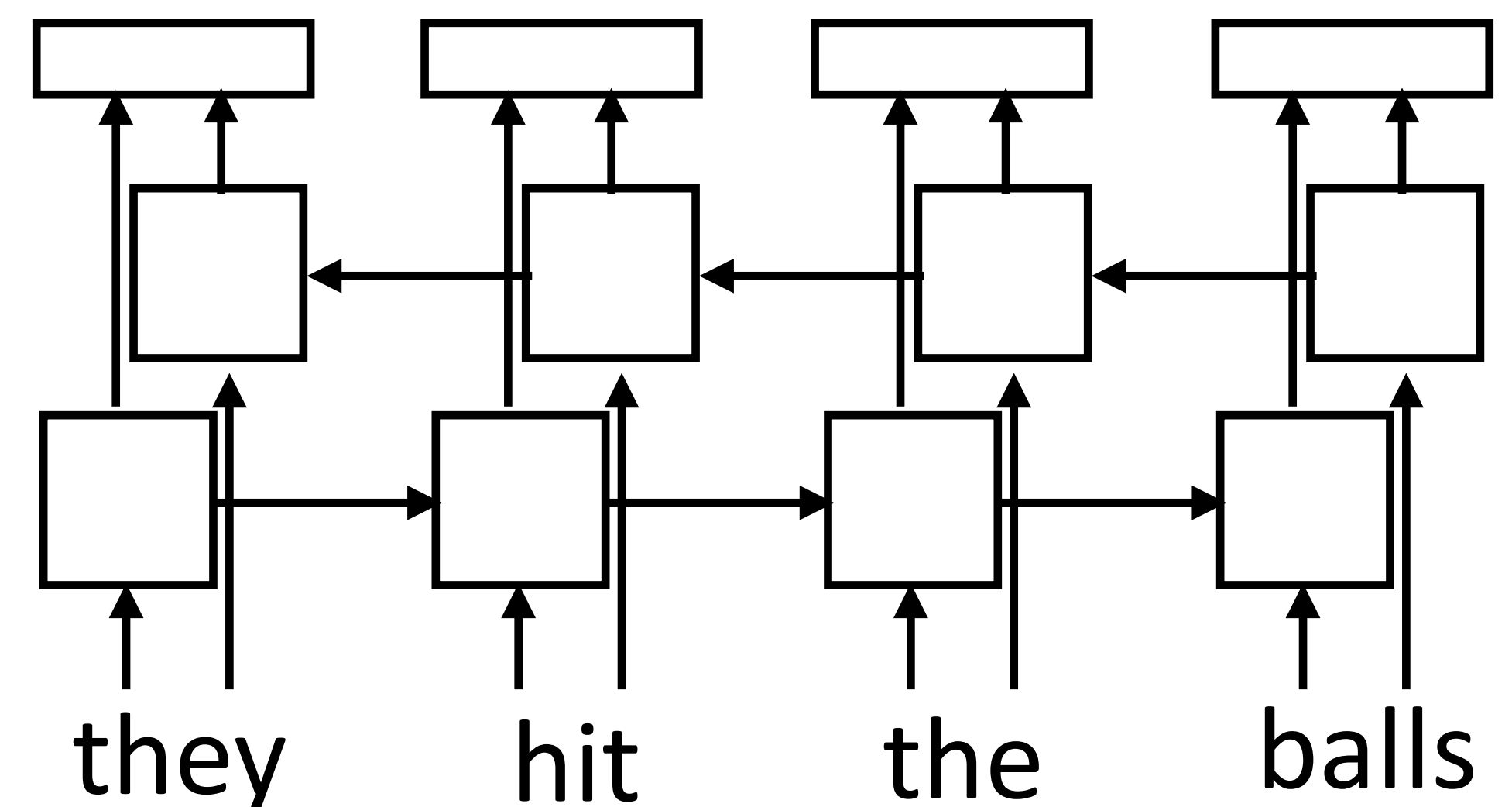
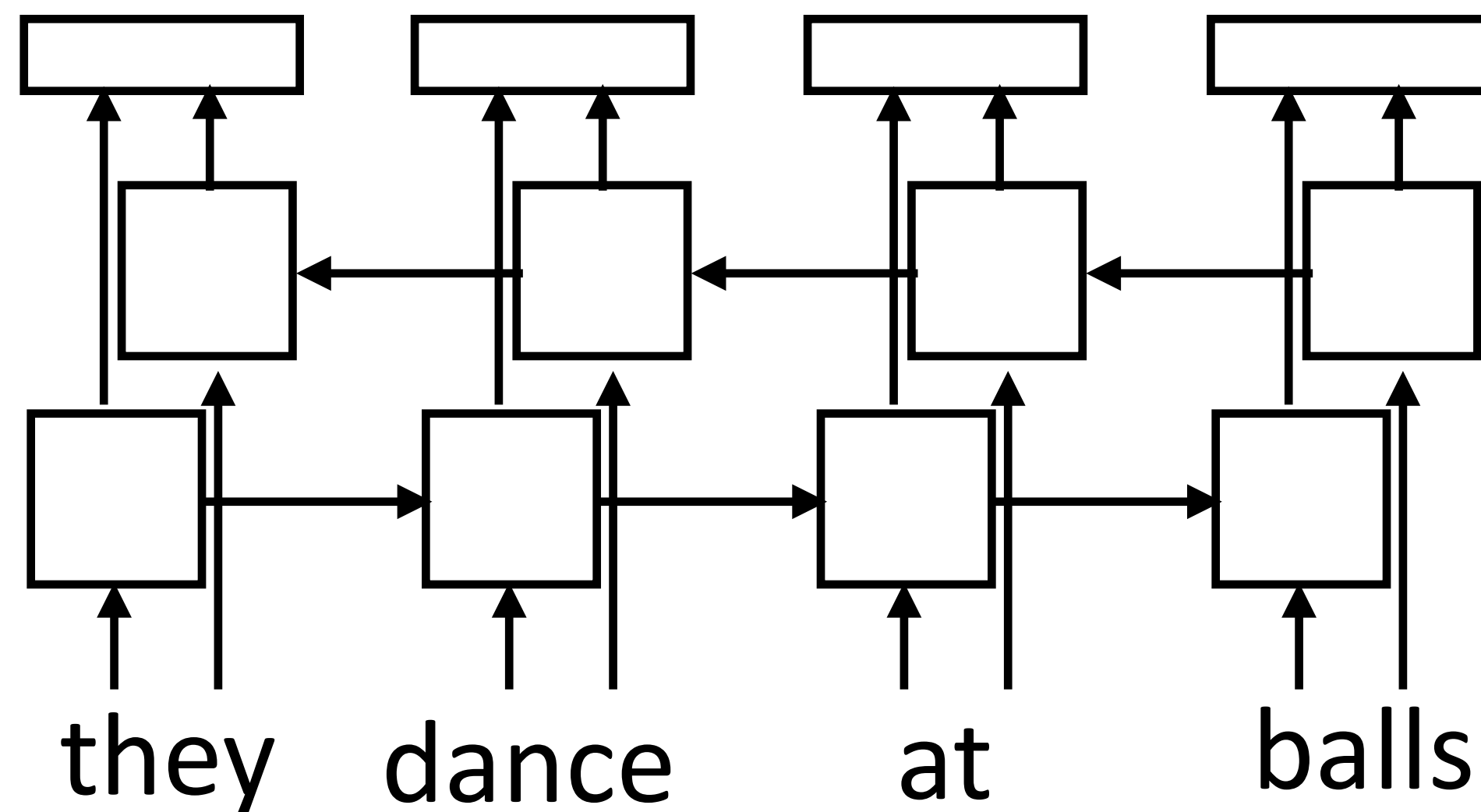
Pre-trained word representations (Mikolov et al., 2013; Pennington et al., 2014) are a key compo-

guage model (LM) objective on a large text corpus. For this reason, we call them ELMo (Embeddings from Language Models) representations. Unlike previous approaches for learning contextualized word vectors (Peters et al., 2017; McCann et al., 2017), ELMo representations are deep, in the sense that they are a function of all of the internal layers of the biLM. More specifically, we learn a linear combination of the vectors stacked above each input word for each end task, which markedly improves performance over just using the top LSTM layer.

Combining the internal states in this manner allows for very rich word representations. Using intrinsic evaluations, we show that the higher-level LSTM states capture context-dependent aspects of word meaning (e.g., they can be used without modification to perform well on supervised word sense disambiguation tasks) while lower-level states model aspects of syntax (e.g., they can be used to do part-of-speech tagging). Simultaneously exposing all of these signals is highly bene-

Context-dependent Embeddings

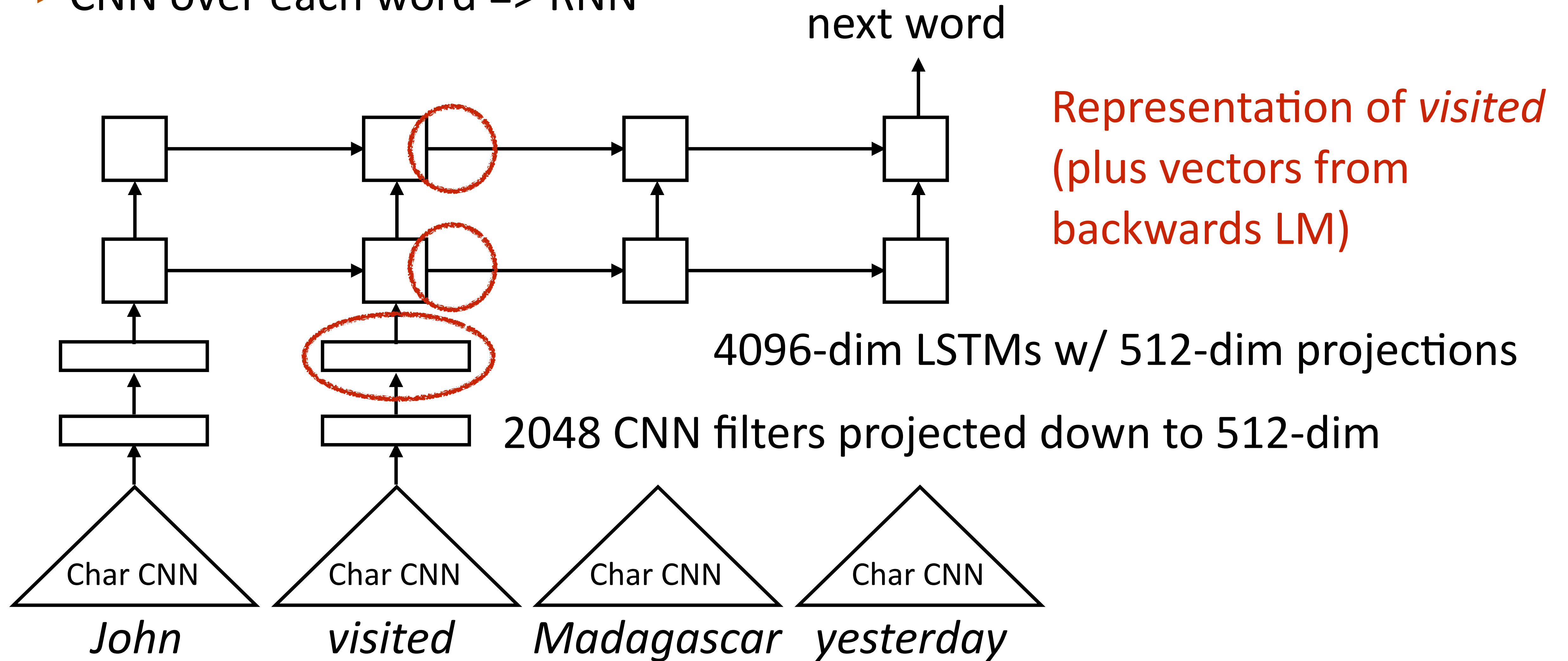
- ▶ How to handle different word senses? One vector for *balls*



- ▶ Train a neural language model to predict the next word given previous words in the sentence, use its internal representations as word vectors
 - ▶ *Context-sensitive* word embeddings: depend on rest of the sentence
 - ▶ *Huge* improvements across nearly all NLP tasks over word2vec & GloVe
- ELMo - Peters et al. (2018)

ELMo

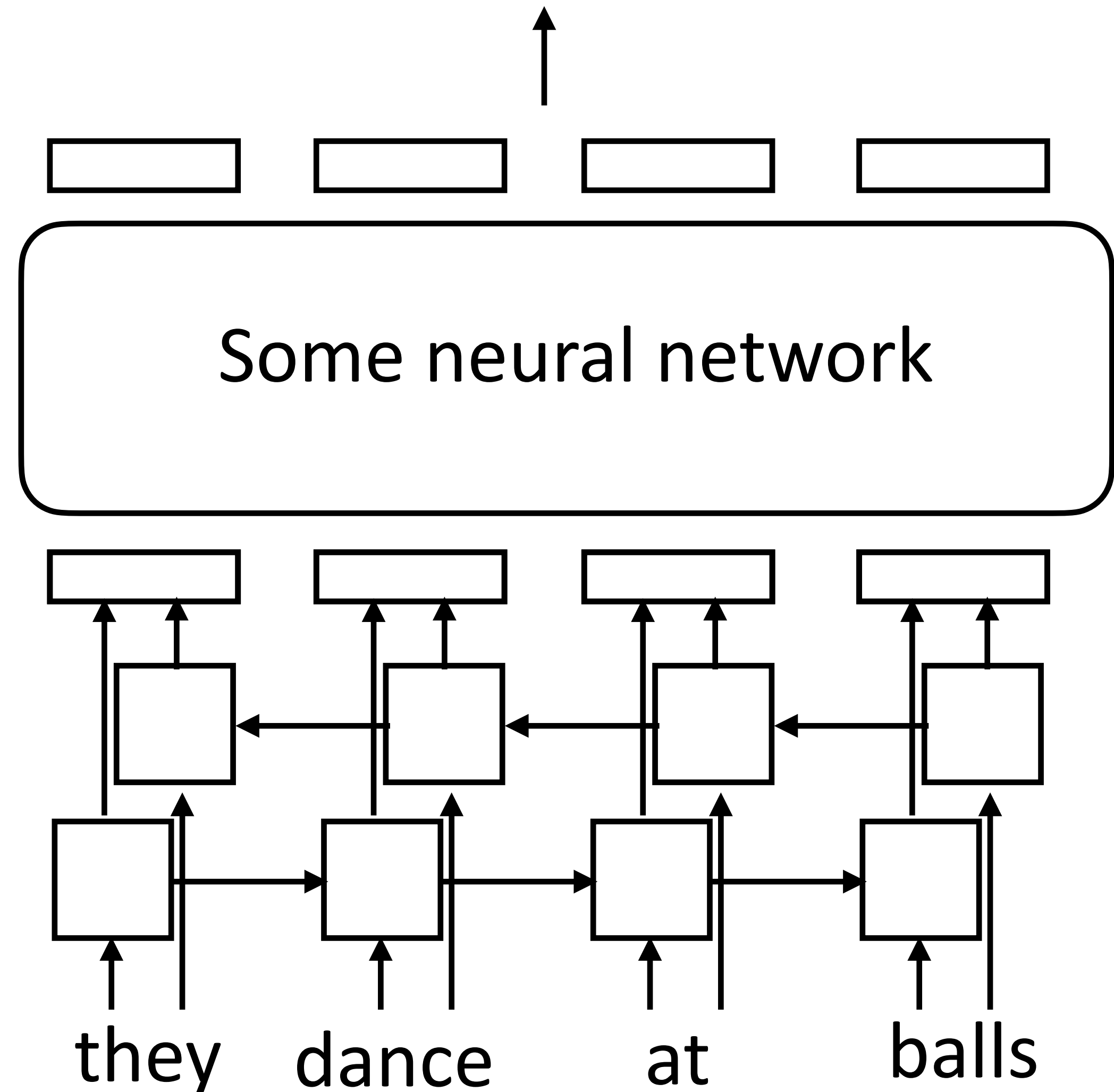
- ▶ CNN over each word => RNN



How to apply ELMo?

- ▶ Take those embeddings and feed them into whatever architecture you want to use for your task
- ▶ *Frozen* embeddings: update the weights of your network but keep ELMo's parameters frozen
- ▶ *Fine-tuning*: backpropagate all the way into ELMo when training your model

Task predictions (sentiment, etc.)



Results: Frozen ELMo

TASK	PREVIOUS SOTA		OUR BASELINE	ELMo + BASELINE	INCREASE (ABSOLUTE/ RELATIVE)
SQuAD	Liu et al. (2017)	84.4	81.1	85.8	4.7 / 24.9%
SNLI	Chen et al. (2017)	88.6	88.0	88.7 $\pm$ 0.17	0.7 / 5.8%
SRL	He et al. (2017)	81.7	81.4	84.6	3.2 / 17.2%
Coref	Lee et al. (2017)	67.2	67.2	70.4	3.2 / 9.8%
NER	Peters et al. (2017)	91.93 $\pm$ 0.19	90.15	92.22 $\pm$ 0.10	2.06 / 21%
SST-5	McCann et al. (2017)	53.7	51.4	54.7 $\pm$ 0.5	3.3 / 6.8%

- Massive improvements across 5 benchmark datasets: question answering, natural language inference, semantic role labeling, coreference resolution, named entity recognition, and sentiment analysis

BERT

BERT

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Jacob Devlin Ming-Wei Chang Kenton Lee Kristina Toutanova

Google AI Language

{jacobdevlin, mingweichang, kentonl, kristout}@google.com

Abstract

We introduce a new language representation model called **BERT**, which stands for **B**idirectional **E**ncoder **R**epresentations from **T**ransformers. Unlike recent language representation models (Peters et al., 2018a; Radford et al., 2018), BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers. As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of tasks, such as question answering and language inference, without substantial task-specific architecture modifications.

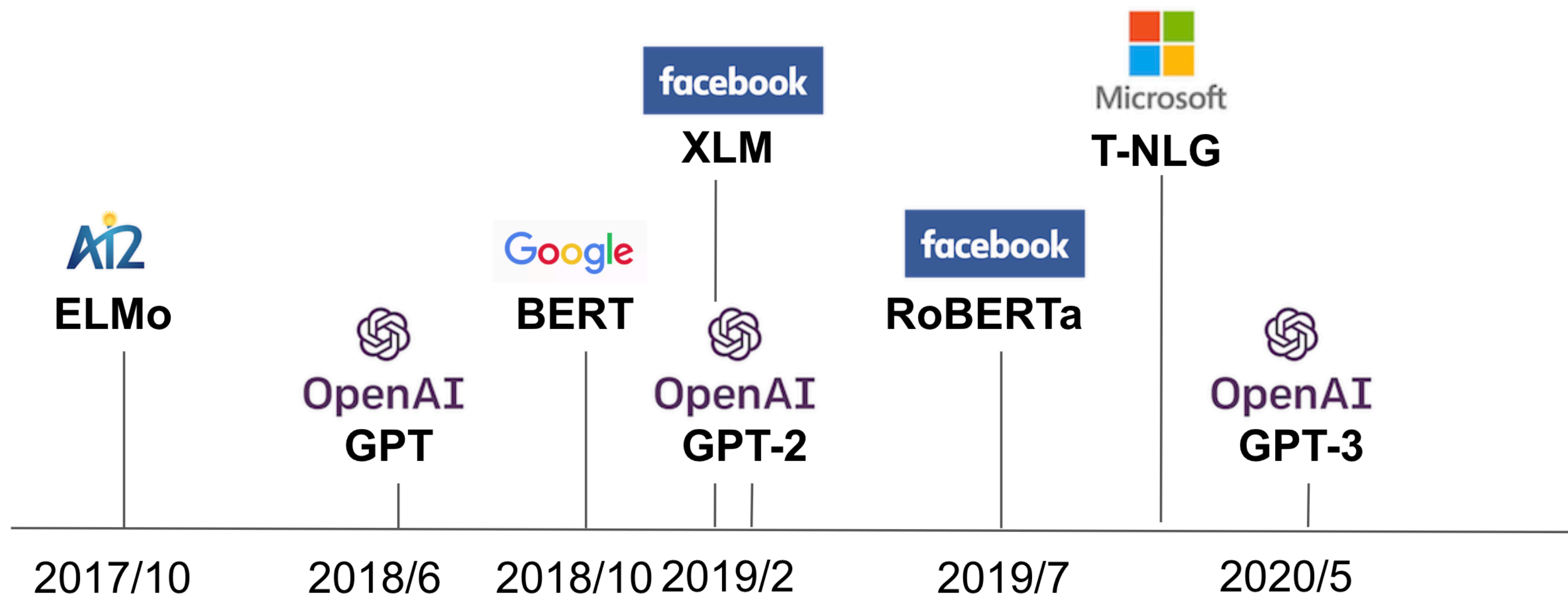
BERT is conceptually simple and empirically powerful. It obtains new state-of-the-art results on eleven natural language processing tasks, including pushing the GLUE score to 80.5% (7.7% point absolute improvement), MultiNLI accuracy to 86.7% (4.6% absolute improvement), SQuAD v1.1 question answering Test F1 to 93.2 (1.5 point absolute improvement) and SQuAD v2.0 Test F1 to 83.1 (5.1 point absolute improvement).

There are two existing strategies for applying pre-trained language representations to downstream tasks: *feature-based* and *fine-tuning*. The feature-based approach, such as ELMo (Peters et al., 2018a), uses task-specific architectures that include the pre-trained representations as additional features. The fine-tuning approach, such as the Generative Pre-trained Transformer (OpenAI GPT) (Radford et al., 2018), introduces minimal task-specific parameters, and is trained on the downstream tasks by simply fine-tuning *all* pre-trained parameters. The two approaches share the same objective function during pre-training, where they use unidirectional language models to learn general language representations.

We argue that current techniques restrict the power of the pre-trained representations, especially for the fine-tuning approaches. The major limitation is that standard language models are unidirectional, and this limits the choice of architectures that can be used during pre-training. For example, in OpenAI GPT, the authors use a left-to-right architecture, where every token can only attend to previous tokens in the self-attention layers

Contextual Word Embeddings

- ▶ AI2 released ELMo in spring 2018, GPT (transformer-based) was released in summer 2018, BERT (transformer-based) came out October 2018
- ▶ aka Pre-trained Language Models



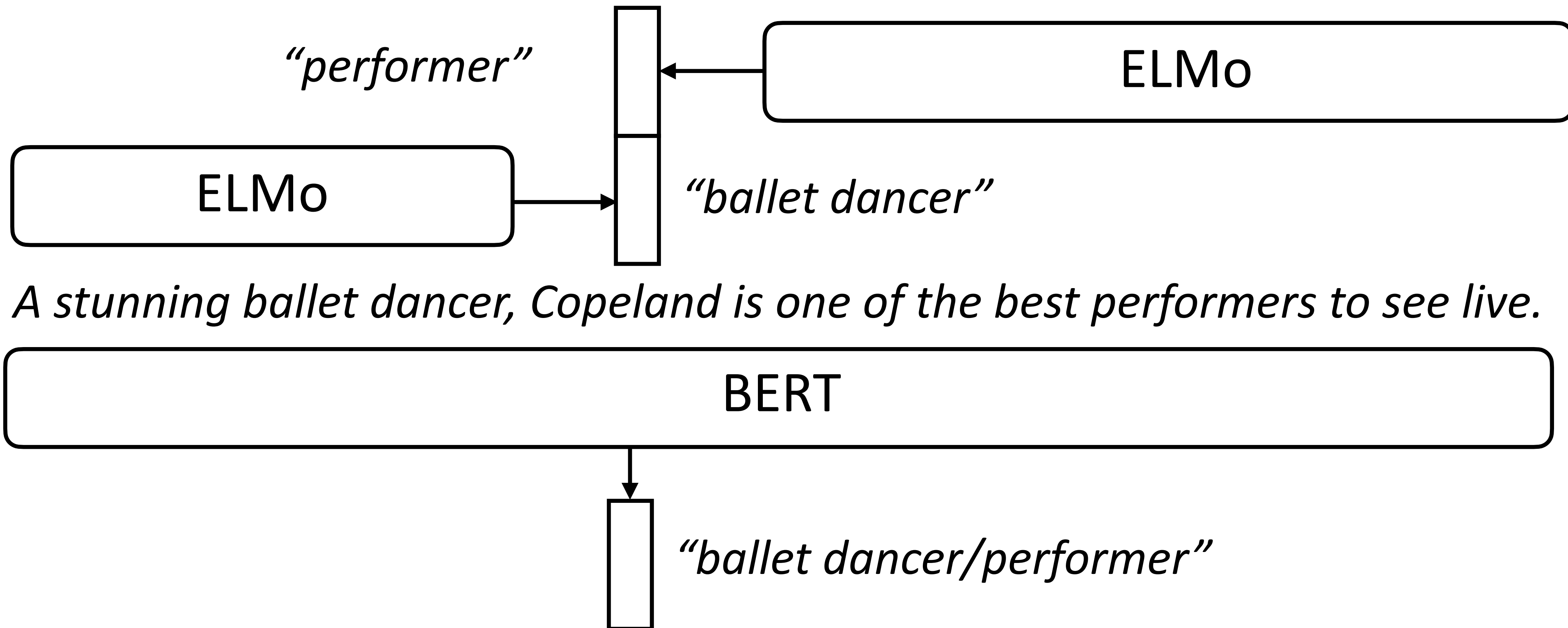
and many more ...

Contextual Word Embeddings

- ▶ AI2 released ELMo in spring 2018, GPT (transformer-based) was released in summer 2018, BERT (transformer-based) came out October 2018
- ▶ aka Pre-trained Language Models
- ▶ BERT's four major changes compared to ELMo:
 - ▶ Transformers instead of LSTMs (transformers in GPT as well)
 - ▶ “Truely” Bidirectional $\Leftrightarrow$ Masked LM objective instead of standard LM
 - ▶ Fine-tune instead of freeze at test time
 - ▶ Uses word pieces (subword tokenization)

BERT

- ELMo is a unidirectional model: we can concatenate two unidirectional models, but is this the right thing to do?

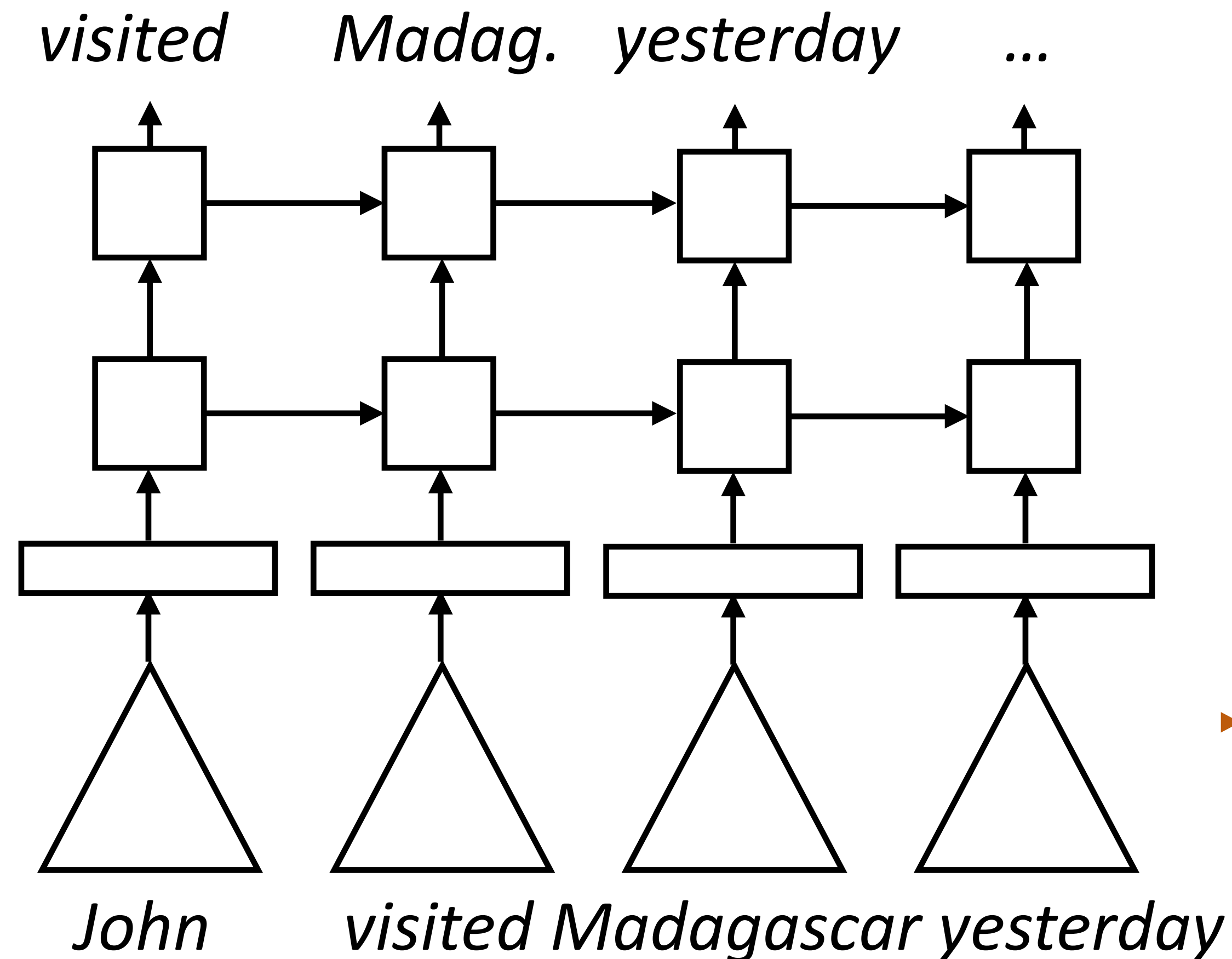


- ELMo looks at each direction in isolation; BERT looks at them jointly
Devlin et al. (2019)

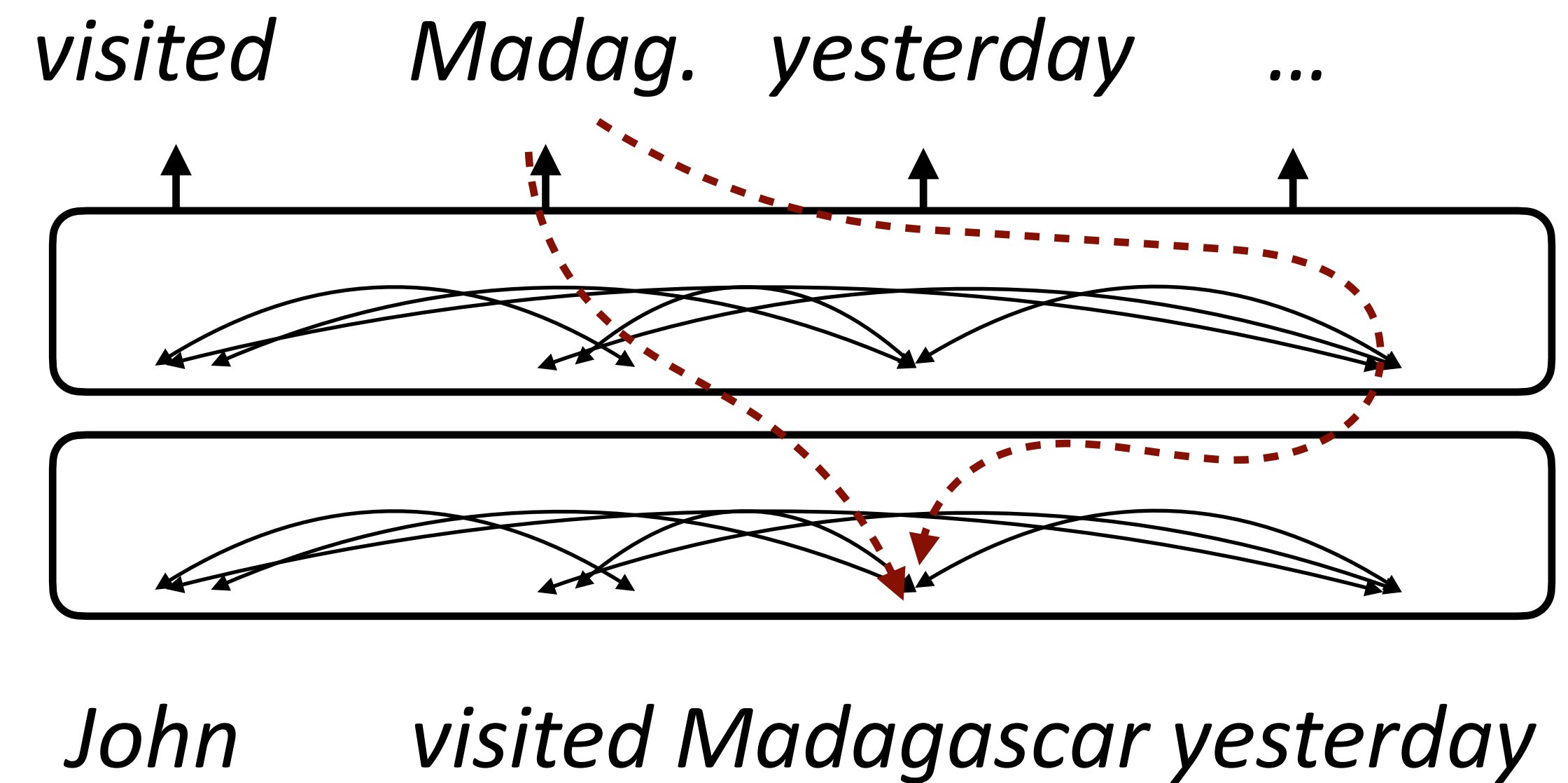
BERT

- How to learn a “deeply bidirectional” model? What happens if we just replace an LSTM with a transformer?

ELMo (Language Modeling)



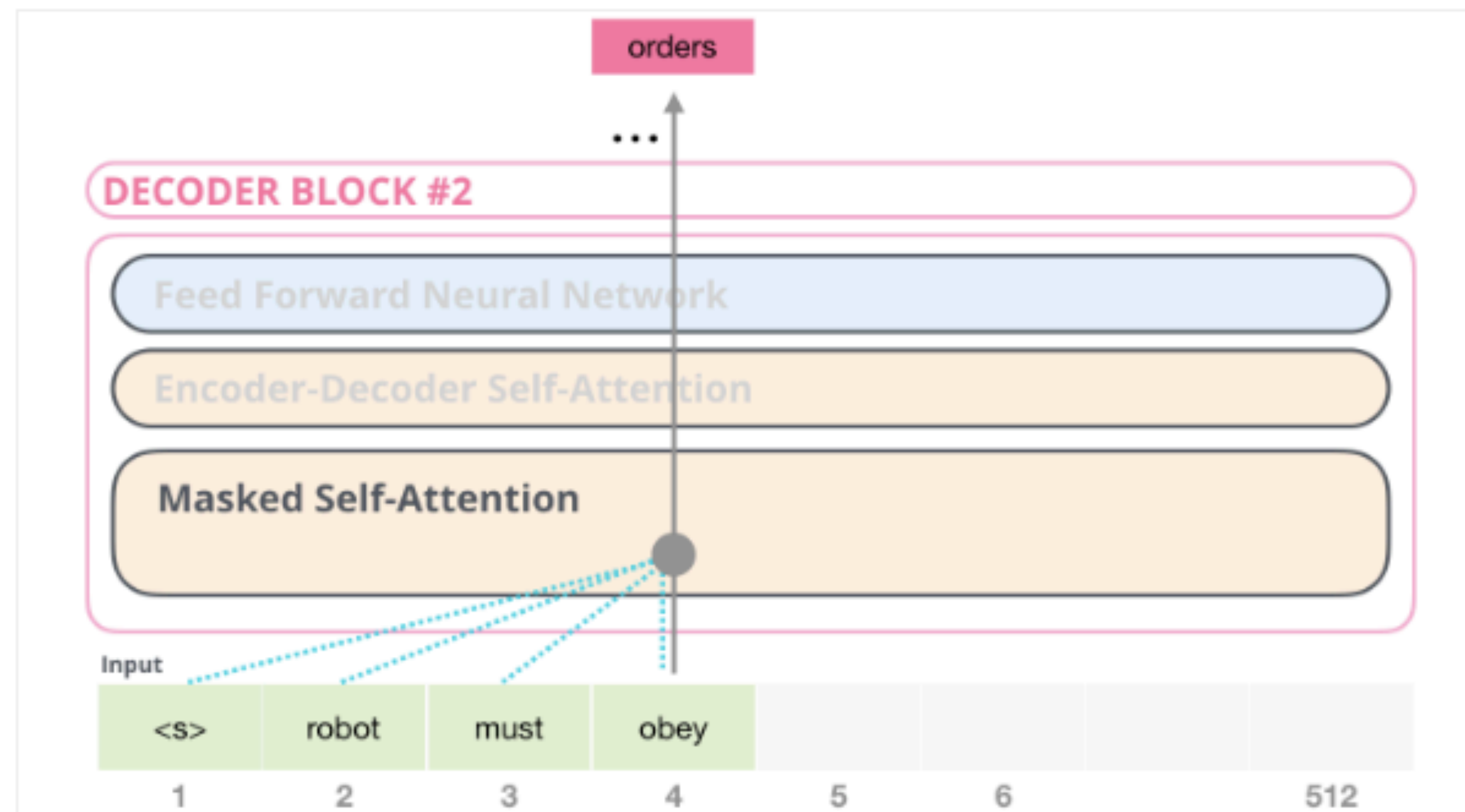
BERT



- Can do this by “one-sided” Transformer (masked self-attention), but “two-sided” Transformer encoder can cheat

GPT

- ▶ Transformer with masked self-attention: each token can only attend to past tokens



Masked Language Modeling (MLM)

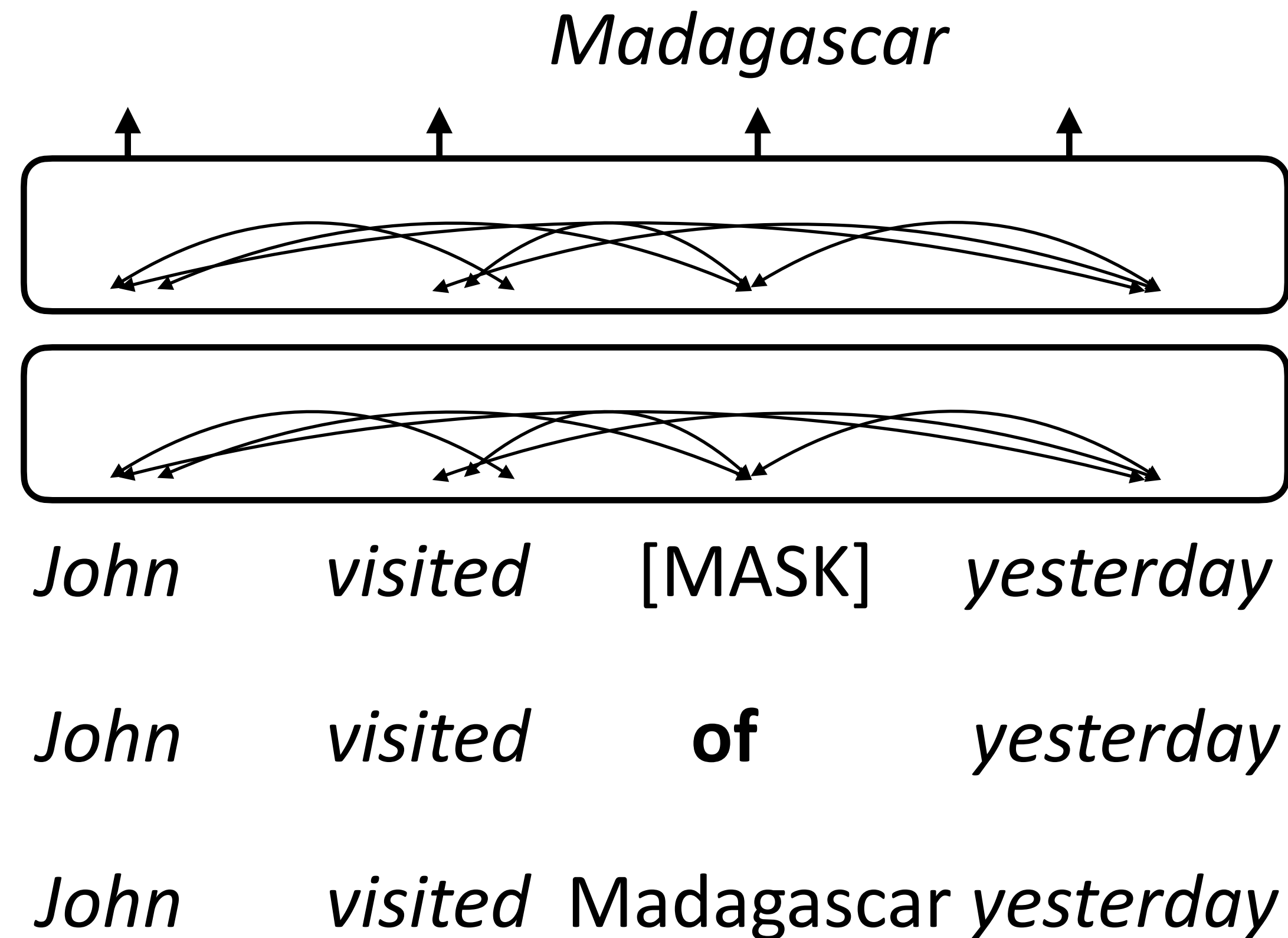
- ▶ How to prevent cheating? Next word prediction fundamentally doesn't work for bidirectional models, instead do *masked language modeling*

- ▶ BERT formula: take a chunk of text, predict 15% of the tokens

- ▶ For 80% (of the 15%), replace the input token with [MASK]

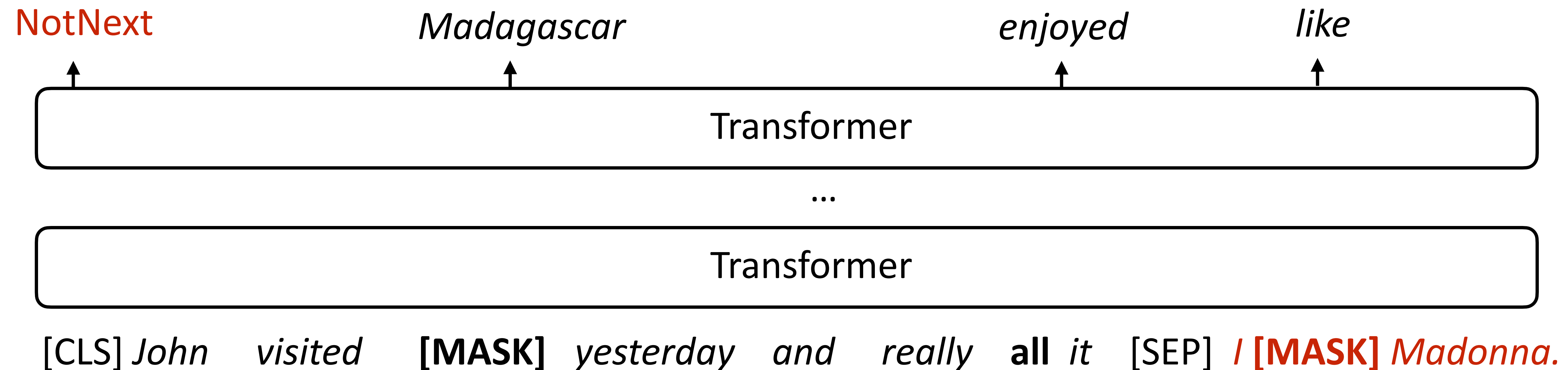
- ▶ For 10%, replace w/random

- ▶ For 10%, keep same



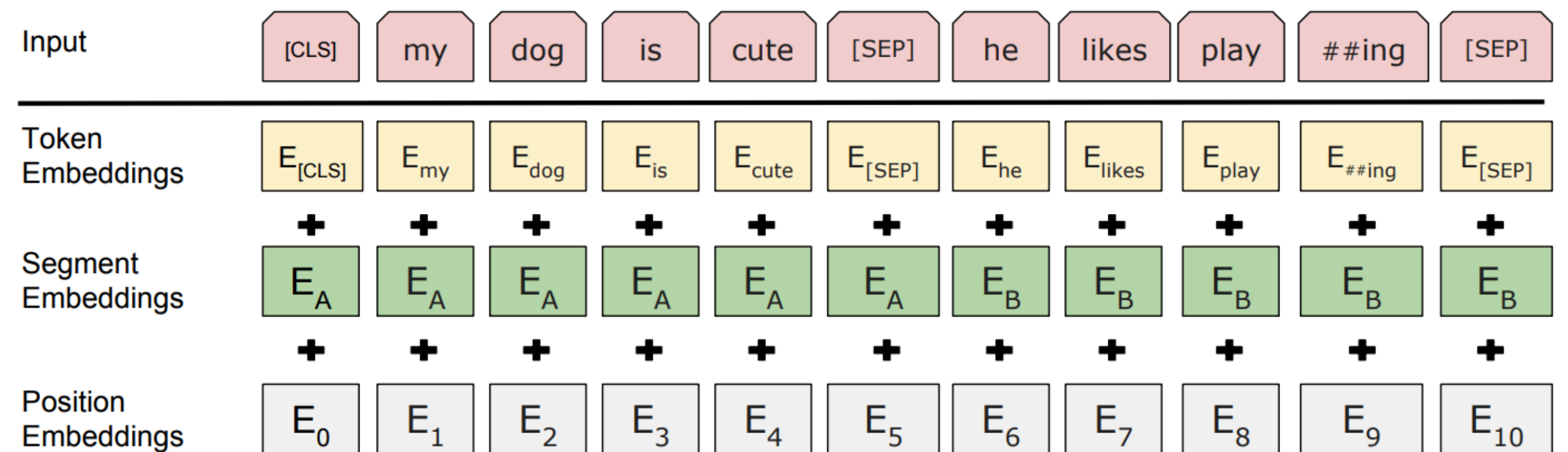
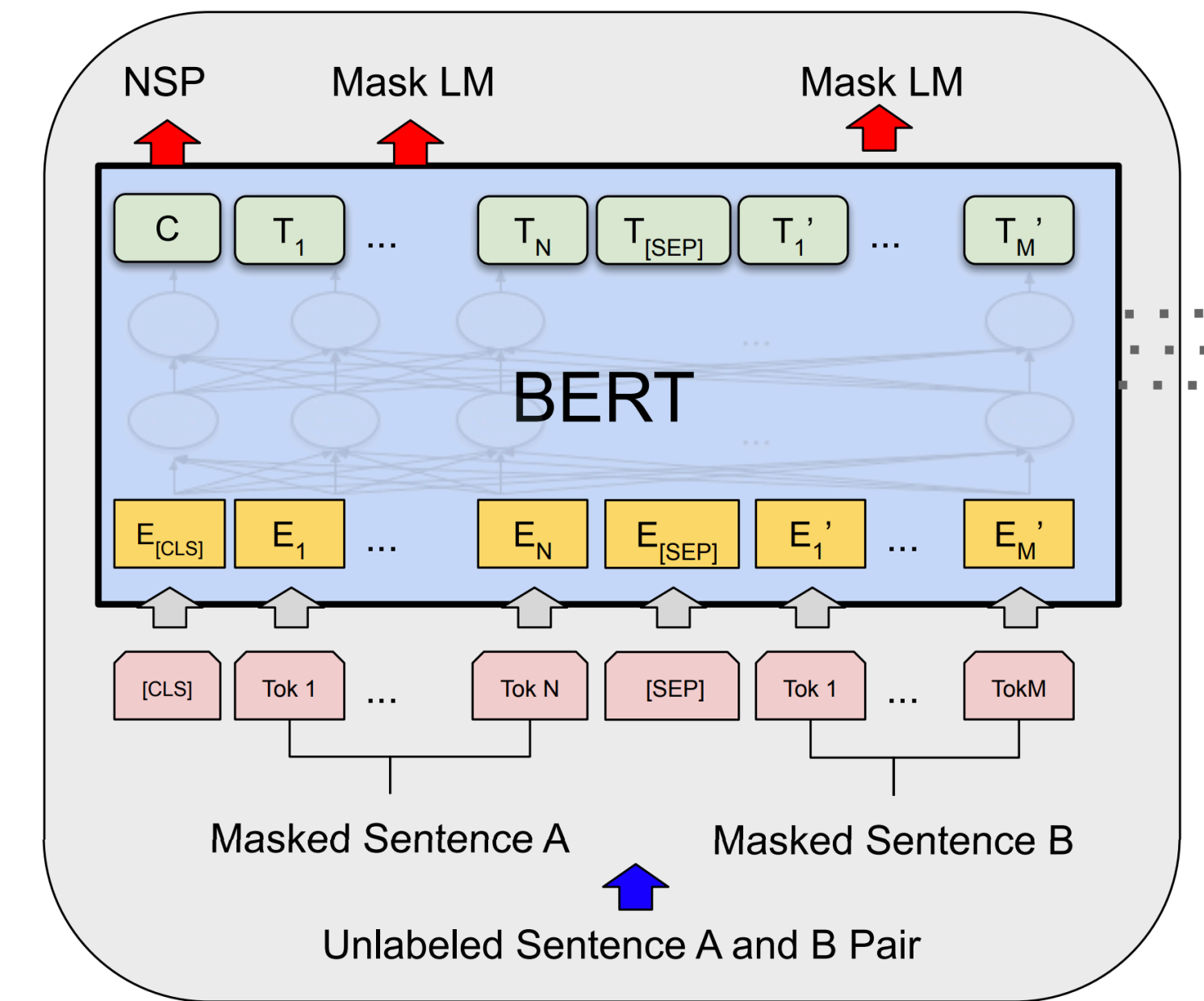
Next “Sentence” Prediction

- ▶ Input: [CLS] Text chunk 1 [SEP] Text chunk 2
- ▶ 50% of the time, take the true next chunk of text, 50% of the time take a random other chunk. Predict whether the next chunk is the “true” next
- ▶ BERT objective: masked LM + next sentence prediction



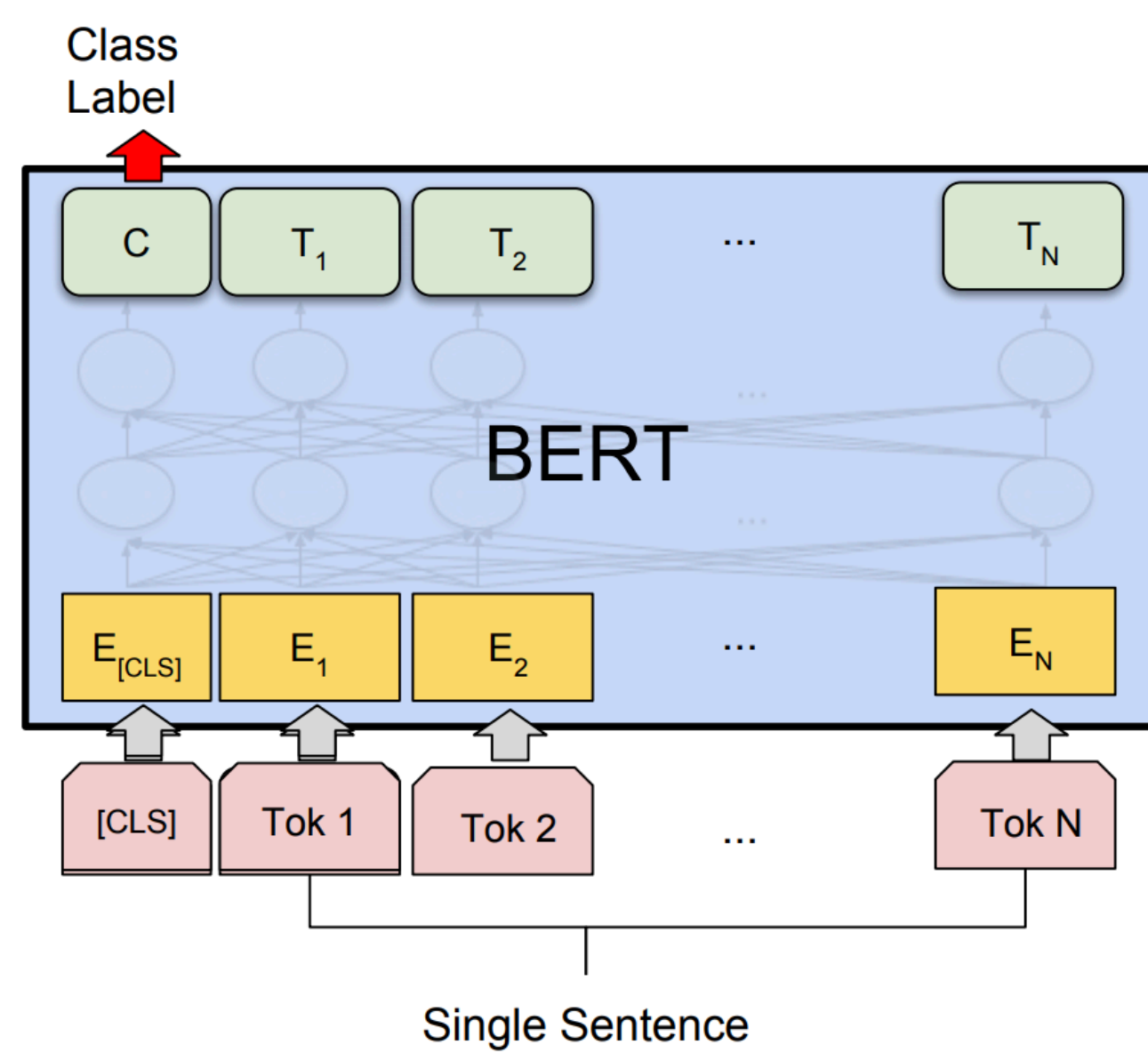
BERT Architecture

- ▶ BERT Base: 12 layers, 768-dim, 12 heads. Total params = 110M
- ▶ BERT Large: 24 layers, 1024-dim, 16 heads. Total params = 340M
- ▶ Positional embeddings and segment embeddings, 30k word pieces
- ▶ This is the model that gets **pre-trained** on a large corpus

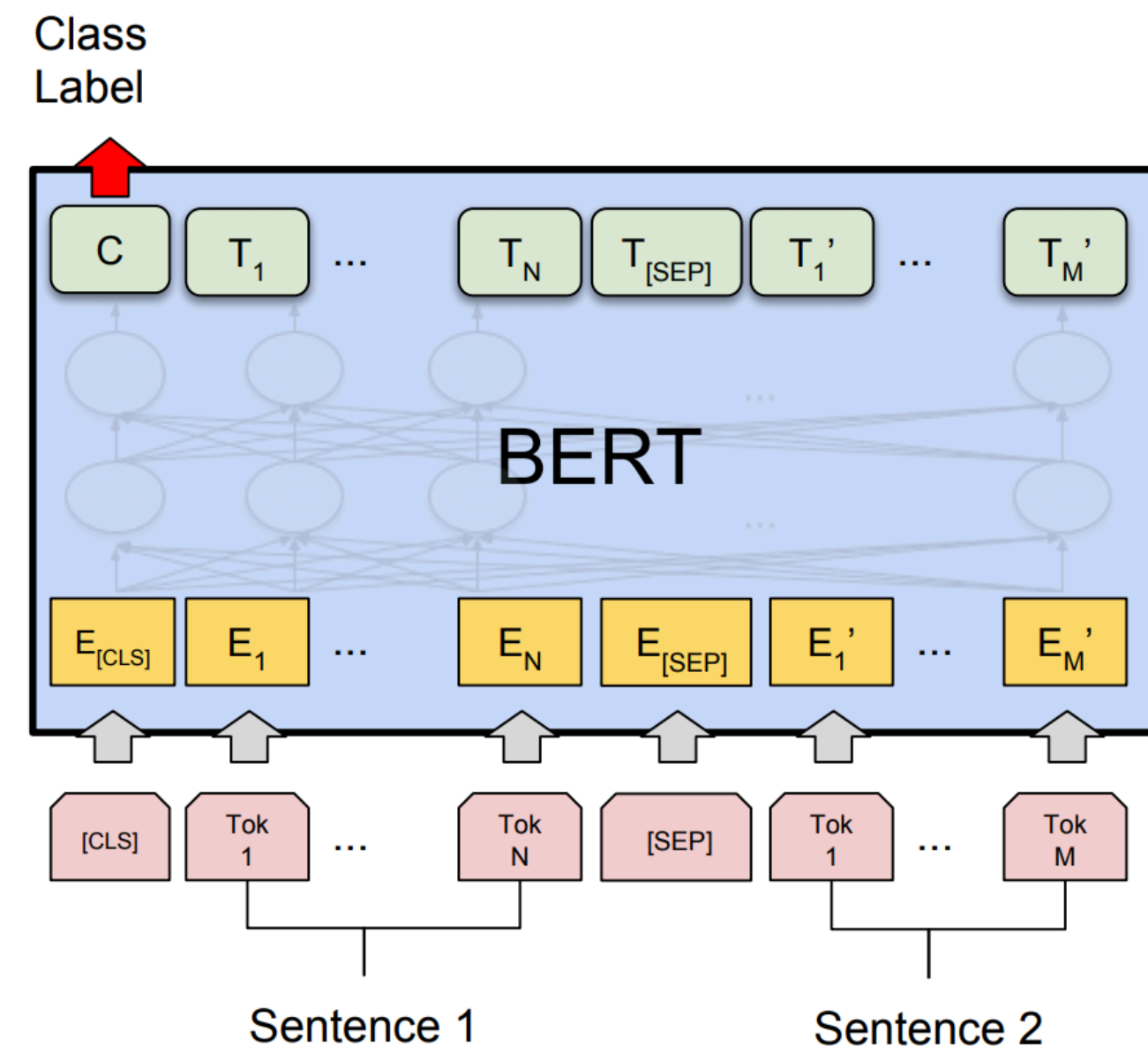


Devlin et al. (2019)

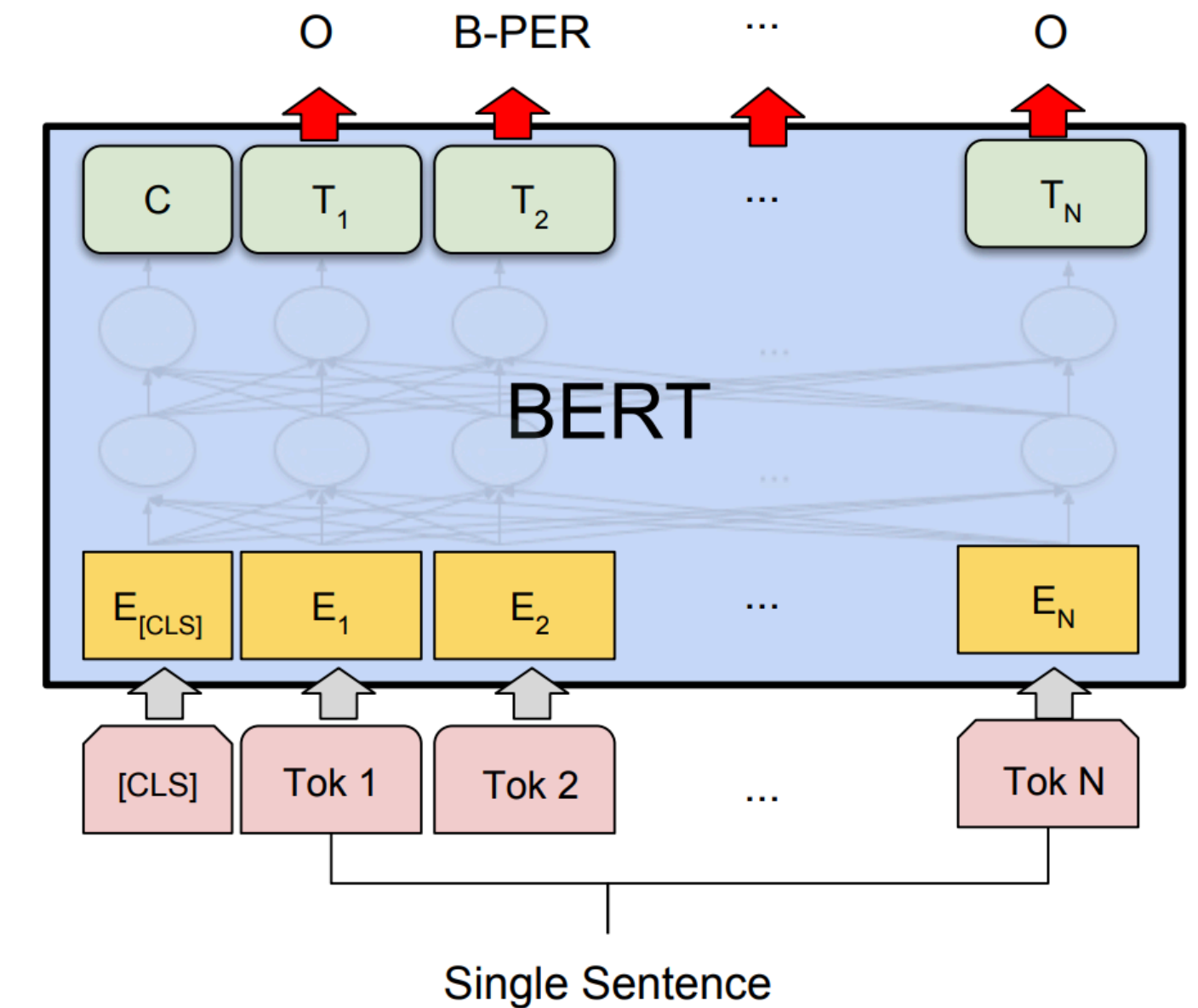
What can BERT do?



(b) Single Sentence Classification Tasks:
SST-2, CoLA



(a) Sentence Pair Classification Tasks:
MNLI, QQP, QNLI, STS-B, MRPC,
RTE, SWAG



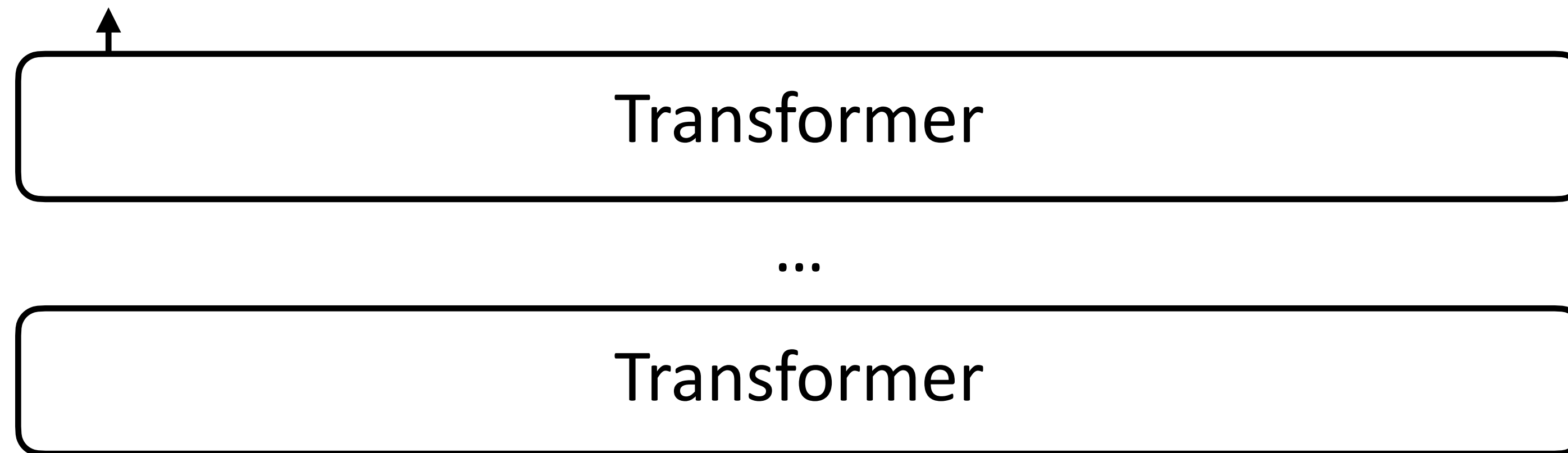
(d) Single Sentence Tagging Tasks:
CoNLL-2003 NER

- ▶ CLS token is used to provide classification decisions
- ▶ Sentence pair tasks (entailment): feed both sentences into BERT
- ▶ BERT can also do tagging by predicting tags at each word piece

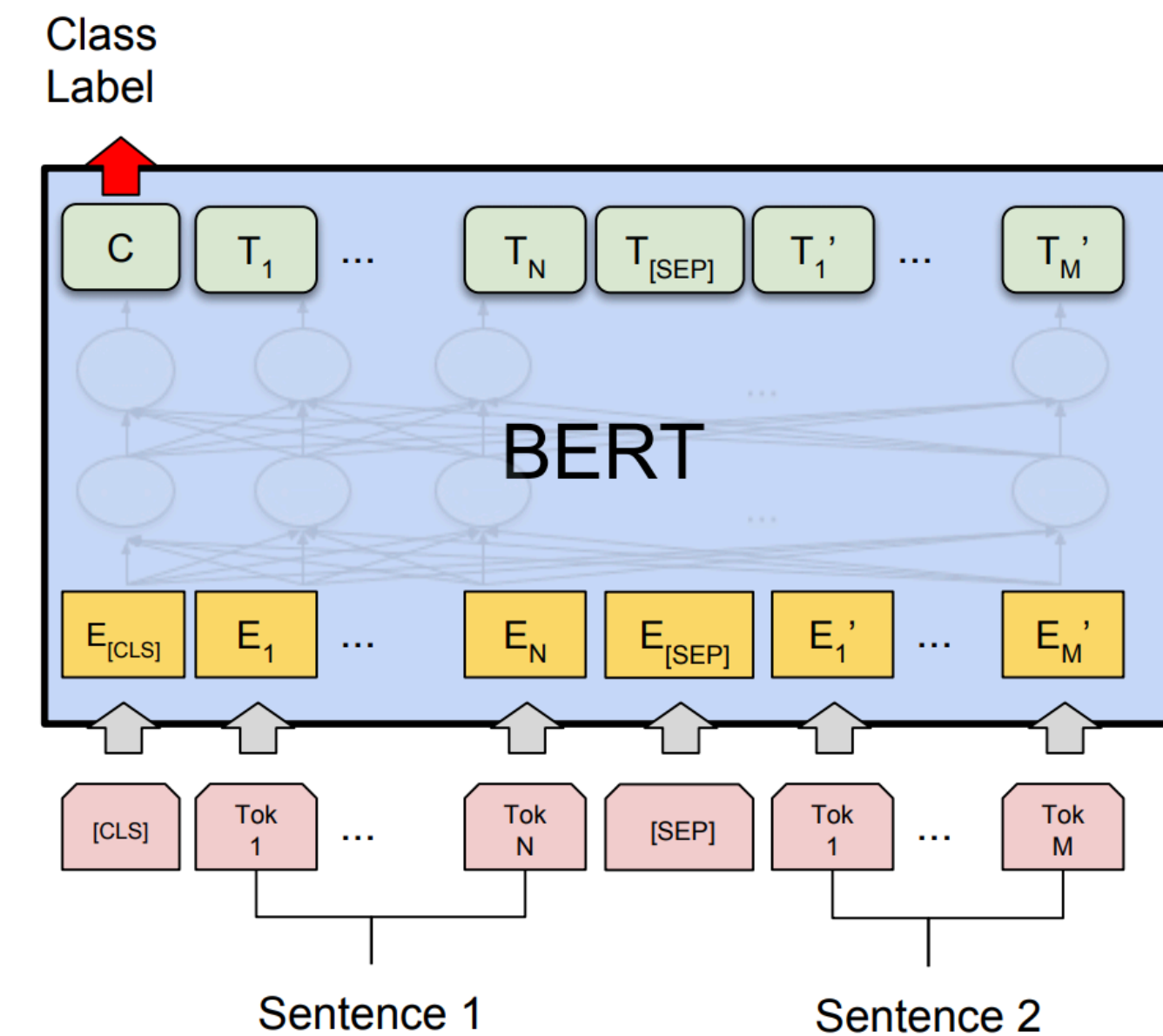
Devlin et al. (2019)

What can BERT do?

Entails (first sentence implies second one is true)



[CLS] A boy plays in the snow [SEP] A boy is outside



(a) Sentence Pair Classification Tasks:
MNLI, QQP, QNLI, STS-B, MRPC,
RTE, SWAG

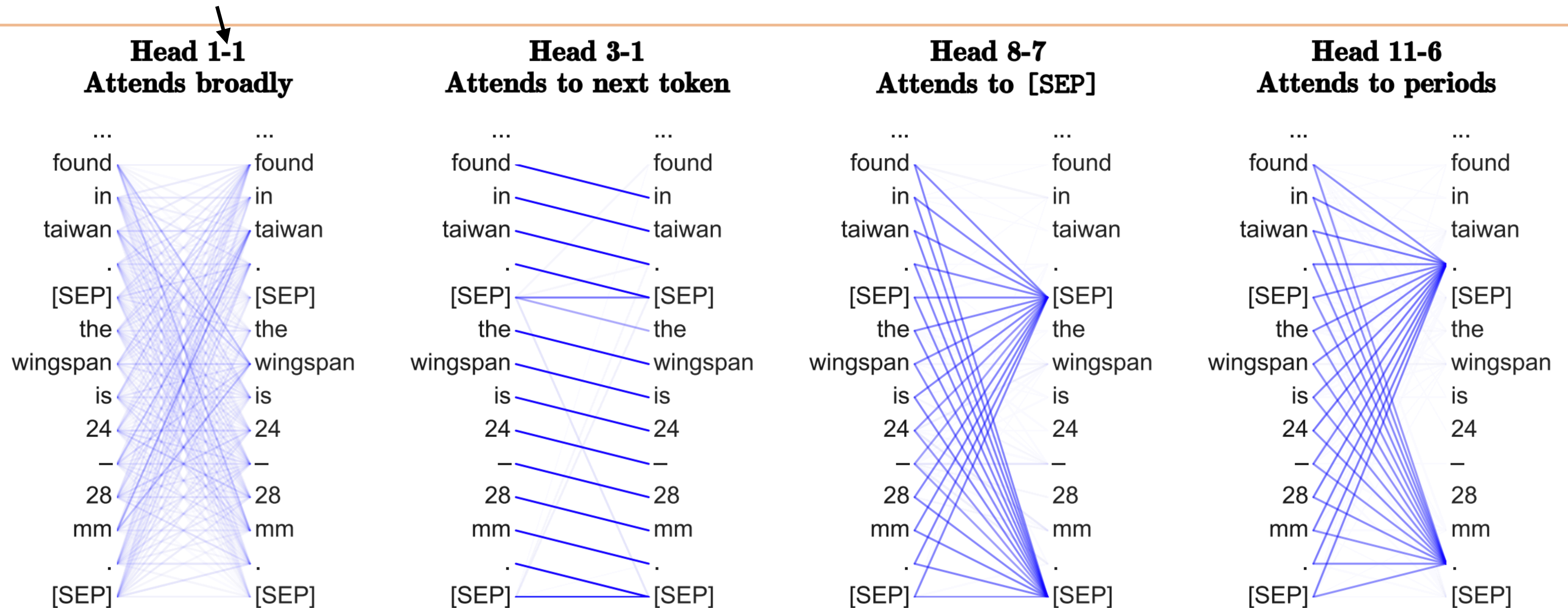
- ▶ How does BERT model this sentence pair stuff?
 - ▶ Transformers can capture interactions between the two sentences, (even though the NSP objective doesn't really cause this to happen).
- Devlin et al. (2019)

What can BERT NOT do?

- ▶ BERT **cannot** generate text (at least not in an obvious way)
 - ▶ Can fill in [MASK] tokens, but can't generate left-to-right (you can put [MASK] at the end, then predict repeatedly, but this is slow)
- ▶ Masked language models are intended to be used primarily for “analysis” tasks, e.g., sequential tagging, semantic similarity between two sentences, ...

What does BERT learn?

layer-head



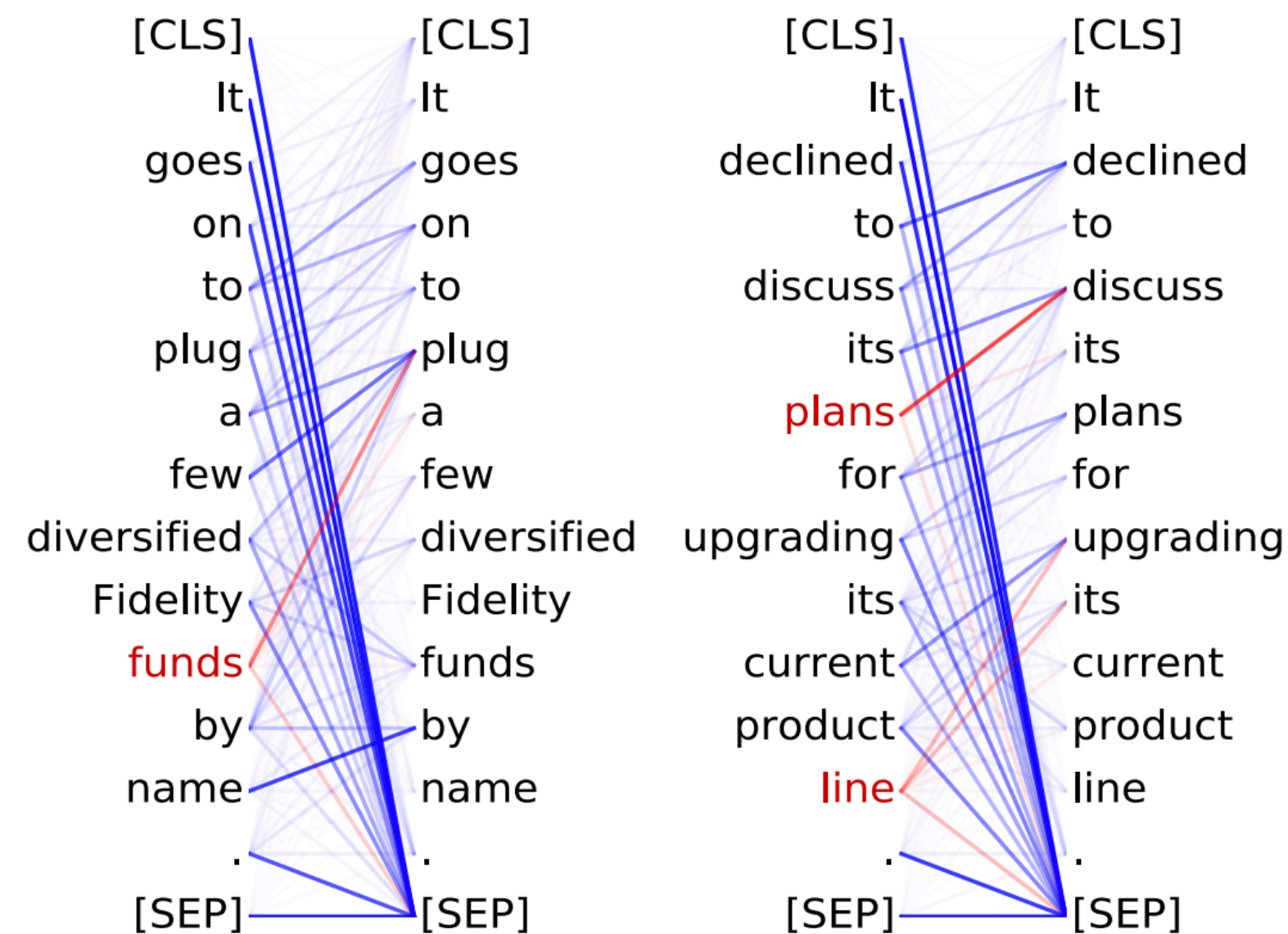
- ▶ Heads on transformers learn interesting and diverse things: content heads (attend based on content), positional heads (based on position), etc.

Clark et al. (2019)

What does BERT learn?

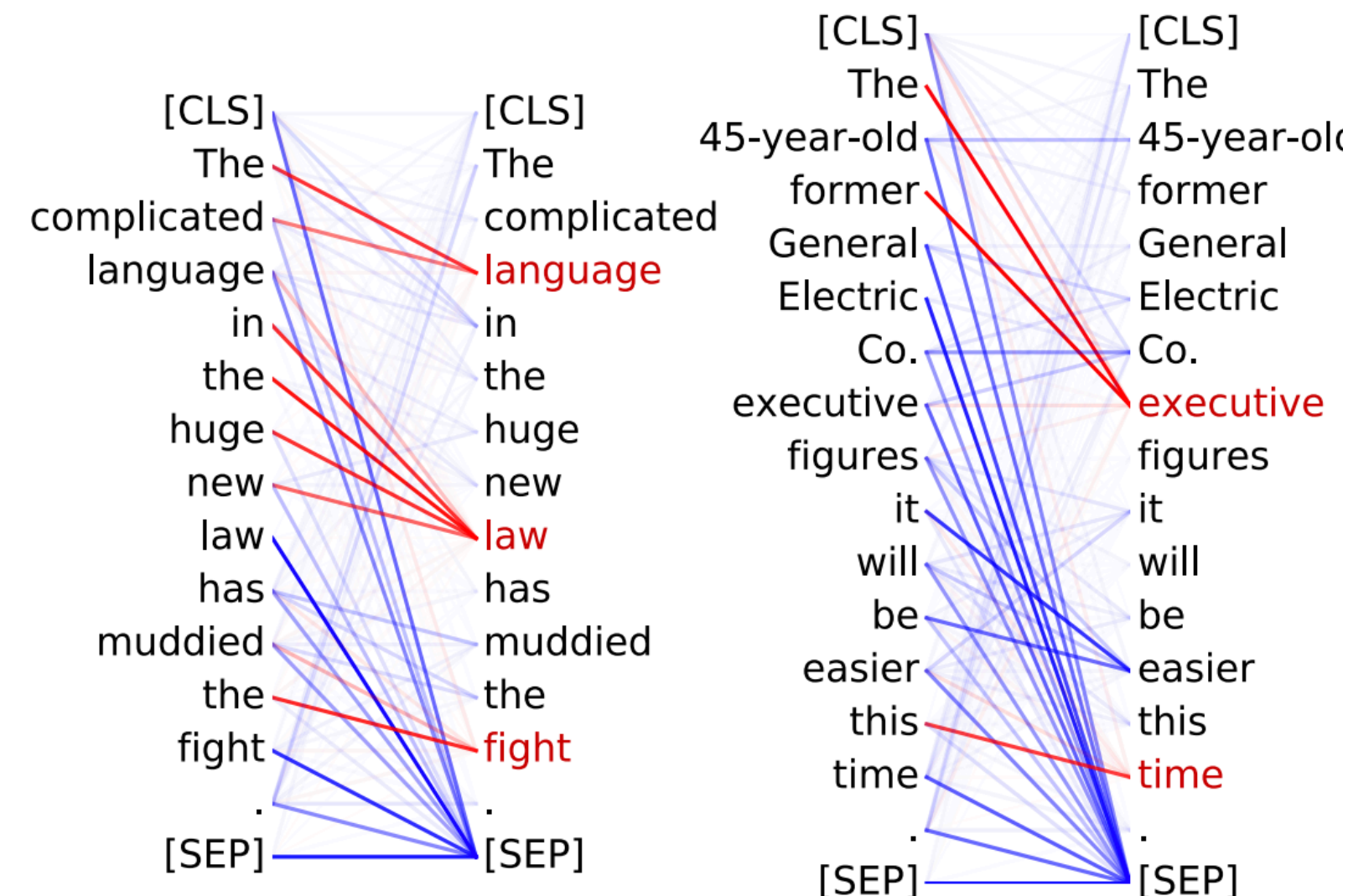
Head 8-10

- **Direct objects** attend to their verbs
- 86.8% accuracy at the dobj relation



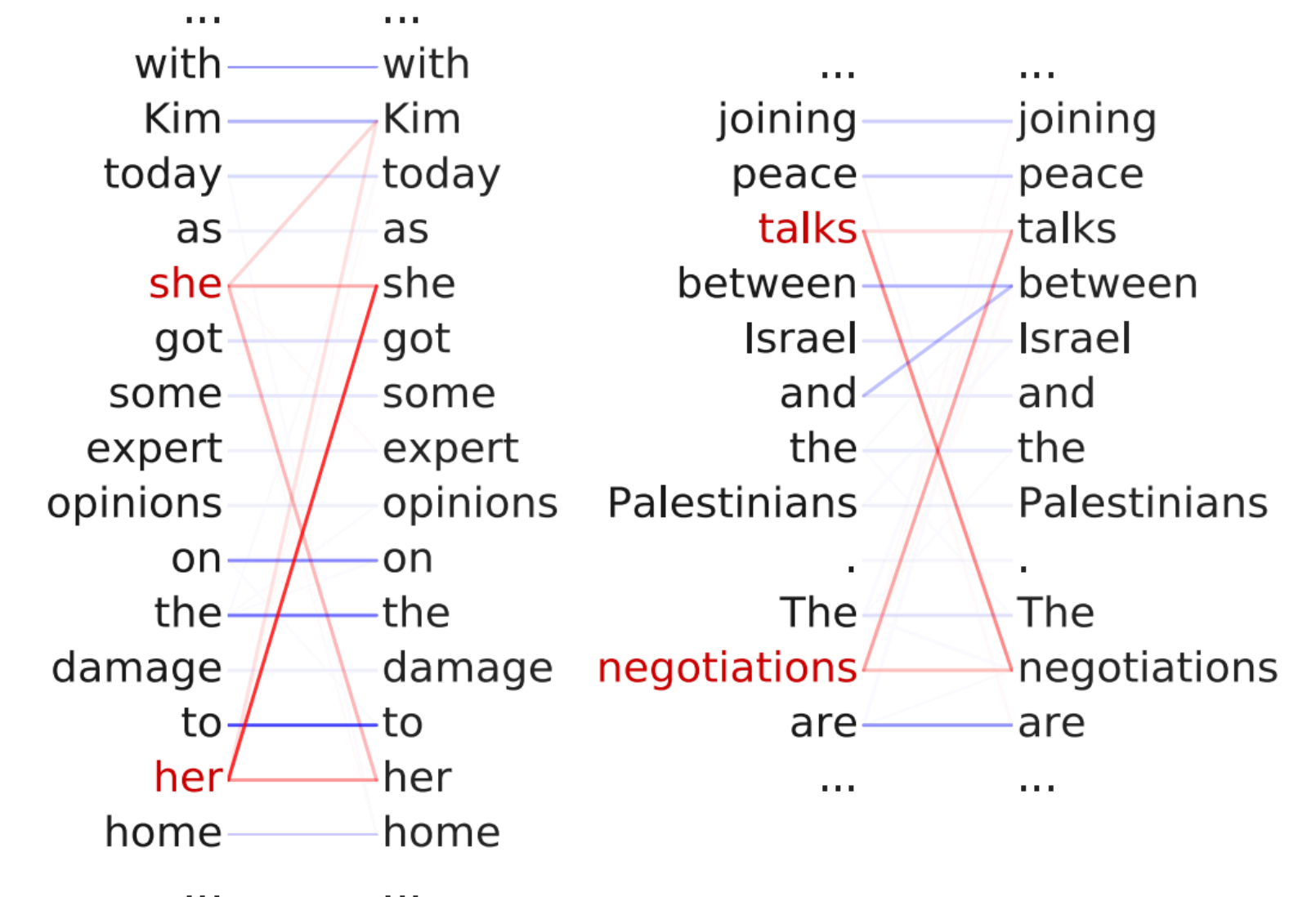
Head 8-11

- **Noun modifiers** (e.g., determiners) attend to their noun
- 94.3% accuracy at the det relation



Head 5-4

- **Coreferent** mentions attend to their antecedents
- 65.1% accuracy at linking the head of a coreferent mention to the head of an antecedent

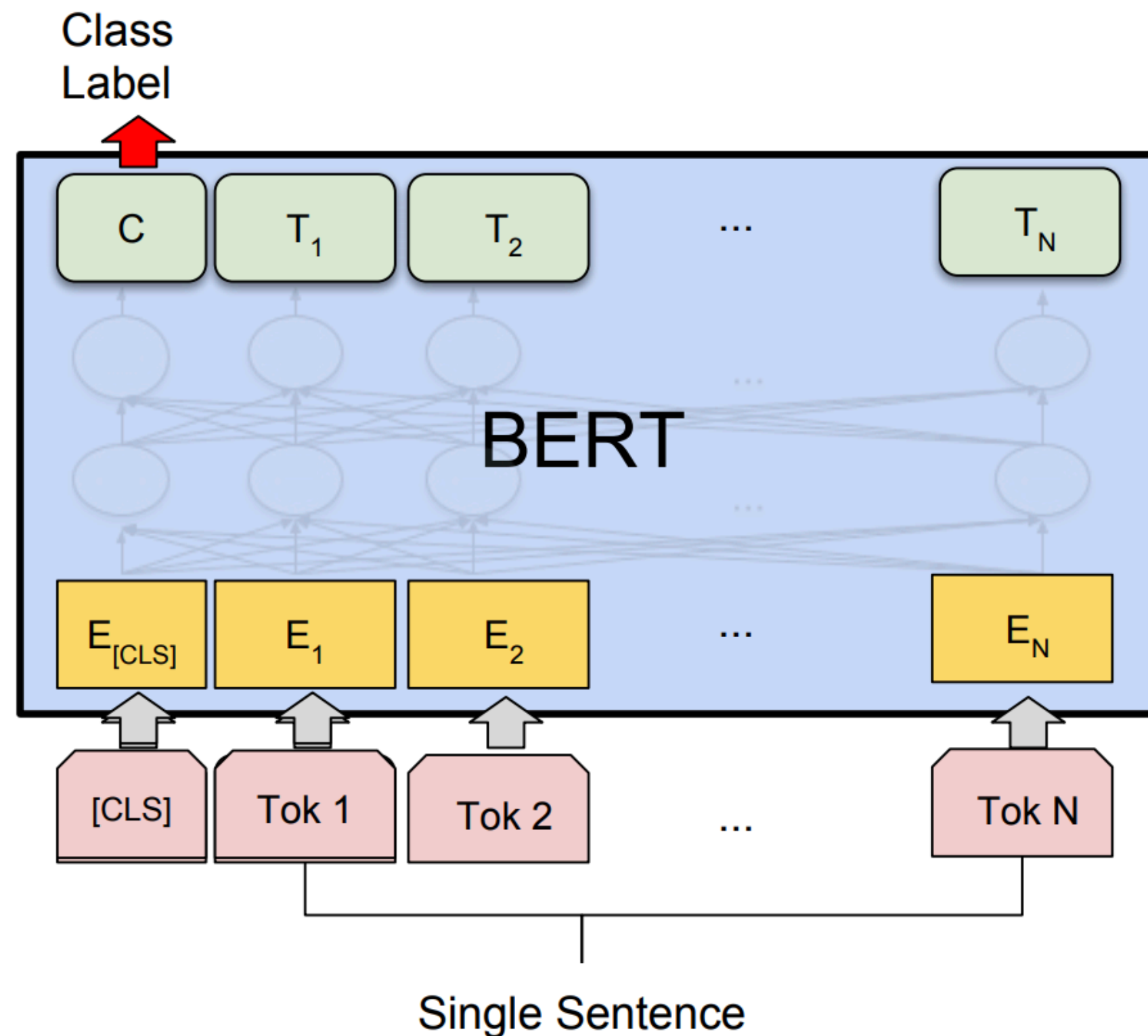


- ▶ Still way worse than what supervised systems can do, but interesting that this is learned organically

BERT Results, Extensions

Fine-tuning BERT

- ▶ Fine-tune for 1-3 epochs, small learning rate (e.g. $2e-5$ - $5e-5$)



(b) Single Sentence Classification Tasks:
SST-2, CoLA

- ▶ Large changes to weights up here (particularly in last layer to route the right information to [CLS])
- ▶ Smaller changes to weights lower down in the transformer
- ▶ Small LR and short fine-tuning schedule mean weights don't change much
- ▶ More complex “triangular learning rate” schemes exist

Fine-tuning BERT

- How does frozen (❄️) vs. fine-tuned (🔥) compare?

Pretraining	Adaptation	NER	SA	Nat. lang. inference		Semantic textual similarity		
		CoNLL 2003	SST-2	MNLI	SICK-E	SICK-R	MRPC	STS-B
Skip-thoughts	❄️	-	81.8	62.9	-	86.6	75.8	71.8
ELMo	❄️	91.7	91.8	79.6	86.3	86.1	76.0	75.9
	🔥	91.9	91.2	76.4	83.3	83.3	74.7	75.5
	$\Delta = \text{🔥} - \text{❄️}$	0.2	-0.6	-3.2	-3.3	-2.8	-1.3	-0.4
BERT-base	❄️	92.2	93.0	84.6	84.8	86.4	78.1	82.9
	🔥	92.4	93.5	84.6	85.8	88.7	84.8	87.1
	$\Delta = \text{🔥} - \text{❄️}$	0.2	0.5	0.0	1.0	2.3	6.7	4.2

- BERT is typically better if the whole network is fine-tuned, unlike ELMo

Evaluation: GLUE

Corpus	Train	Test	Task	Metrics	Domain
Single-Sentence Tasks					
CoLA	8.5k	1k	acceptability	Matthews corr.	misc.
SST-2	67k	1.8k	sentiment	acc.	movie reviews
Similarity and Paraphrase Tasks					
MRPC	3.7k	1.7k	paraphrase	acc./F1	news
STS-B	7k	1.4k	sentence similarity	Pearson/Spearman corr.	misc.
QQP	364k	391k	paraphrase	acc./F1	social QA questions
Inference Tasks					
MNLI	393k	20k	NLI	matched acc./mismatched acc.	misc.
QNLI	105k	5.4k	QA/NLI	acc.	Wikipedia
RTE	2.5k	3k	NLI	acc.	news, Wikipedia
WNLI	634	146	coreference/NLI	acc.	fiction books

Wang et al. (2019)

Results

System	MNLI-(m/mm) 392k	QQP 363k	QNLI 108k	SST-2 67k	CoLA 8.5k	STS-B 5.7k	MRPC 3.5k	RTE 2.5k	Average -
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.9	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	88.1	91.3	45.4	80.0	82.3	56.0	75.2
BERT _{BASE}	84.6/83.4	71.2	90.1	93.5	52.1	85.8	88.9	66.4	79.6
BERT _{LARGE}	86.7/85.9	72.1	91.1	94.9	60.5	86.5	89.3	70.1	81.9

- ▶ Huge improvements over prior work (even compared to ELMo)
- ▶ Effective at “sentence pair” tasks: textual entailment (does sentence A imply sentence B), paraphrase detection

Devlin et al. (2018)

Subsequent Improvements to BERT

- ▶ Dynamic masking: standard BERT uses the same MASK scheme for every epoch, RoBERTa recomputes them

epoch 2

epoch 1

... John visited Madagascar yesterday ...

- ▶ Whole word masking: don't mask out parts of words (word pieces)

... _John _visited _Mada gas car yesterday ...

RoBERTa

- ▶ “Robustly optimized BERT” incorporating some of these tricks

Model	data	bsz	steps	SQuAD (v1.1/2.0)	MNLI-m	SST-2
RoBERTa						
with BOOKS + WIKI	16GB	8K	100K	93.6/87.3	89.0	95.3
+ additional data (§3.2)	160GB	8K	100K	94.0/87.7	89.3	95.6
+ pretrain longer	160GB	8K	300K	94.4/88.7	90.0	96.1
+ pretrain even longer	160GB	8K	500K	94.6/89.4	90.2	96.4
BERT _{LARGE}						
with BOOKS + WIKI	13GB	256	1M	90.9/81.8	86.6	93.7

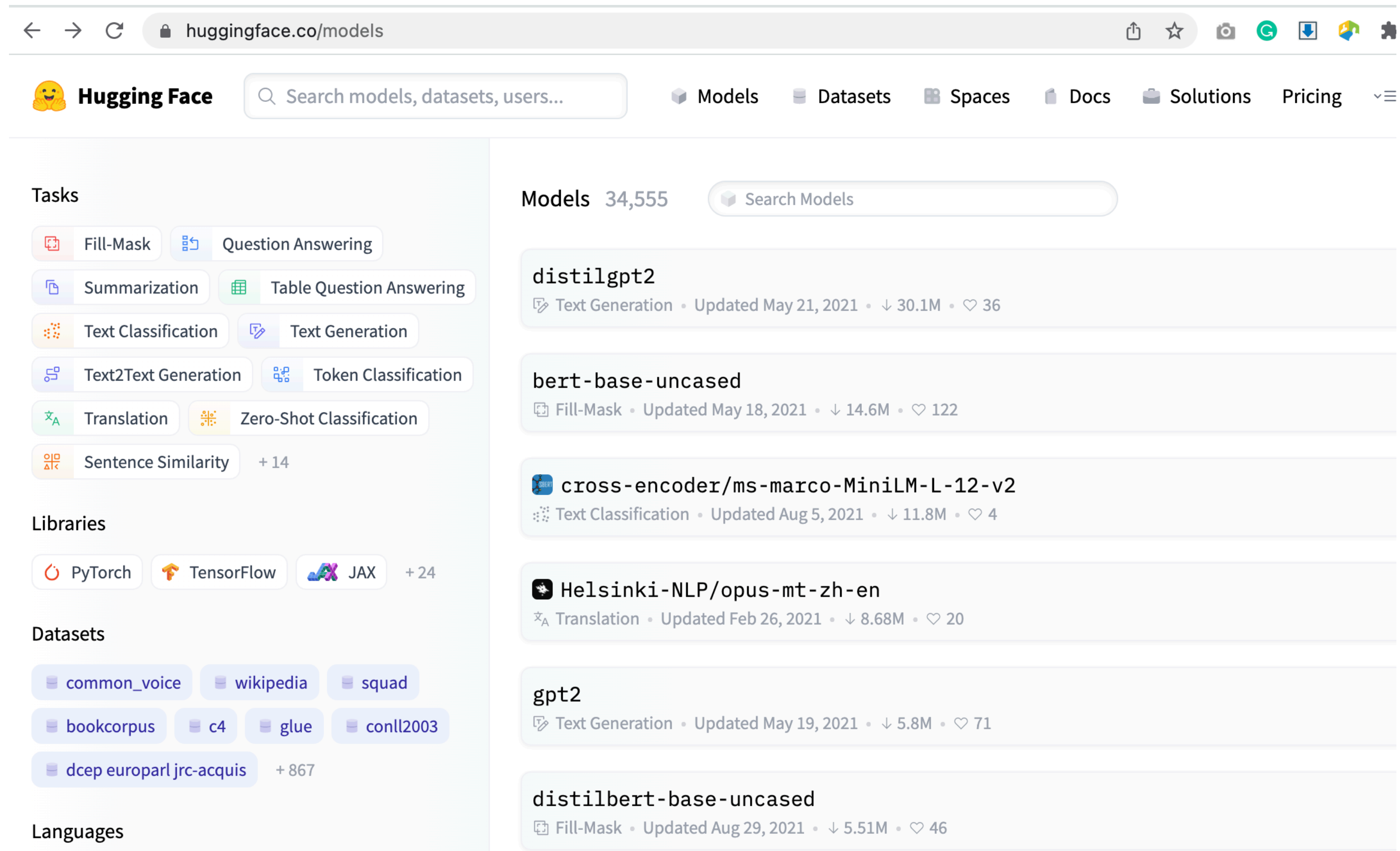
- ▶ 160GB of data instead of 16 GB

- ▶ New training + more data = better performance

- ▶ For this and more: check out Huggingface or fairseq

many BERT variations

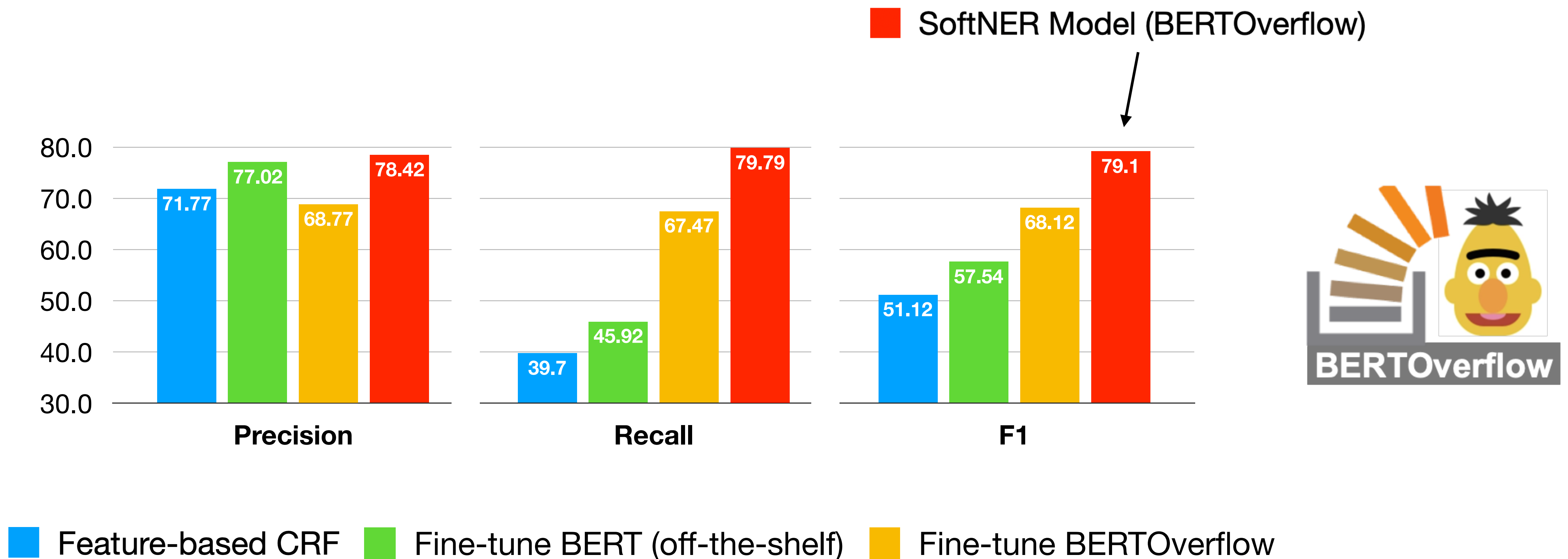
- For specific text domains (e.g. StackOverflow), or specific languages



The screenshot shows the Hugging Face website's models page. The browser address bar displays 'huggingface.co/models'. The page features a navigation bar with the Hugging Face logo, a search bar, and links to Models, Datasets, Spaces, Docs, Solutions, and Pricing. On the left sidebar, there are sections for Tasks (Fill-Mask, Question Answering, Summarization, Table Question Answering, Text Classification, Text Generation, Text2Text Generation, Token Classification, Translation, Zero-Shot Classification, Sentence Similarity), Libraries (PyTorch, TensorFlow, JAX), Datasets (common_voice, wikipedia, squad, bookcorpus, c4, glue, conll2003, dcep europarl jrc-acquis), and Languages. The main content area shows a list of models under the heading 'Models 34,555'. The first five models listed are: distilgpt2 (Text Generation, Updated May 21, 2021, 30.1M, 36 likes), bert-base-uncased (Fill-Mask, Updated May 18, 2021, 14.6M, 122 likes), cross-encoder/ms-marco-MiniLM-L-12-v2 (Text Classification, Updated Aug 5, 2021, 11.8M, 4 likes), Helsinki-NLP/opus-mt-zh-en (Translation, Updated Feb 26, 2021, 8.68M, 20 likes), and gpt2 (Text Generation, Updated May 19, 2021, 5.8M, 71 likes). The bottom model shown is distilbert-base-uncased (Fill-Mask, Updated Aug 29, 2021, 5.51M, 46 likes).

NER in StackOverflow

- ▶ A domain-specific BERT model that pre-trained on 10-year StackOverflow data (152M sentences; ~2B tokens).



Multilingual BERT

Bilingual BERT

- ▶ A customized bilingual BERT for Arabic NLP and English-to-Arabic zero-shot transfer learning

	Data Source	Data Size (All / English / Arabic)	IE Performance (F1 score)
AraBERT (AUBeirut 2019)	News	2.5B / 0B / 2.5B	97.1 / —
mBERT (Google 2018)	Wiki	21.9B / 2.5B / 0.15B	75.3 / 30.1
XLM-RoBERTa (Facebook 2019)	Common Crawl	295B / 55.6B / 2.9B	79.2 / 40.4
GigaBERT (our work)	News, Wiki, Common Crawl	10.4B / 6.1B / 4.3B	84.3 / 48.2

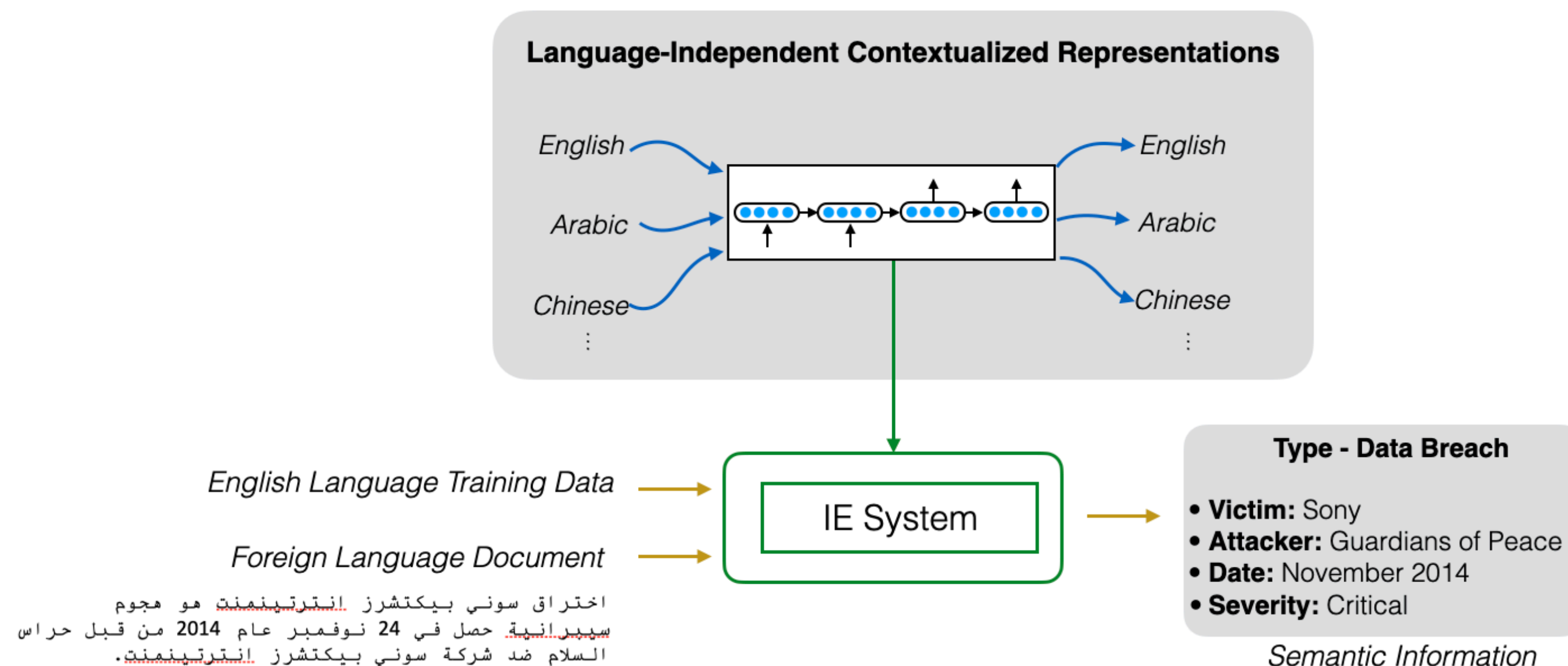
supervised learning

zero-shot transfer learning

Lan et al. (2020)

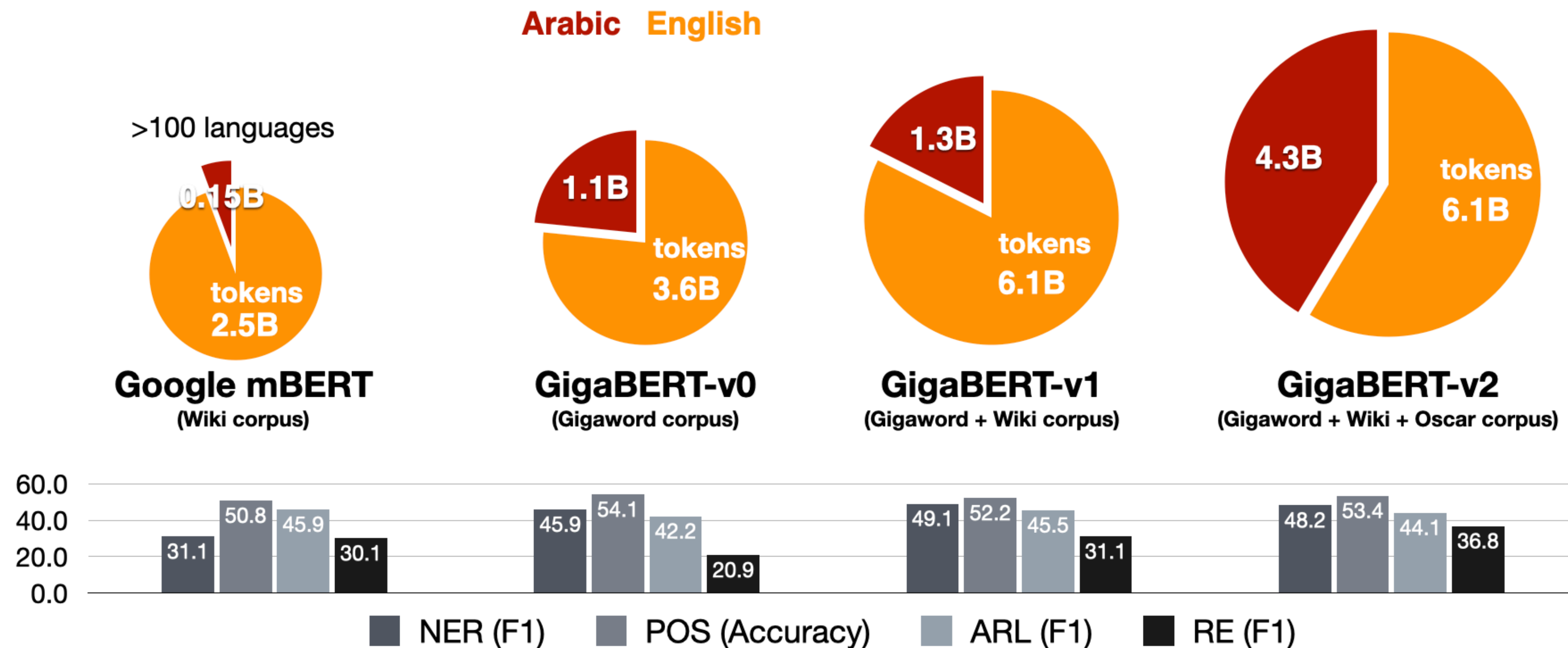
Bilingual BERT

- ▶ A customized bilingual BERT for Arabic NLP and English-to-Arabic zero-shot transfer learning
- ▶ i.e., Information extraction models, trained on annotated English data, directly apply to non-English texts to extract entities and events.



Bilingual BERT

- ▶ A customized bilingual BERT for Arabic NLP and English-to-Arabic zero-shot transfer learning

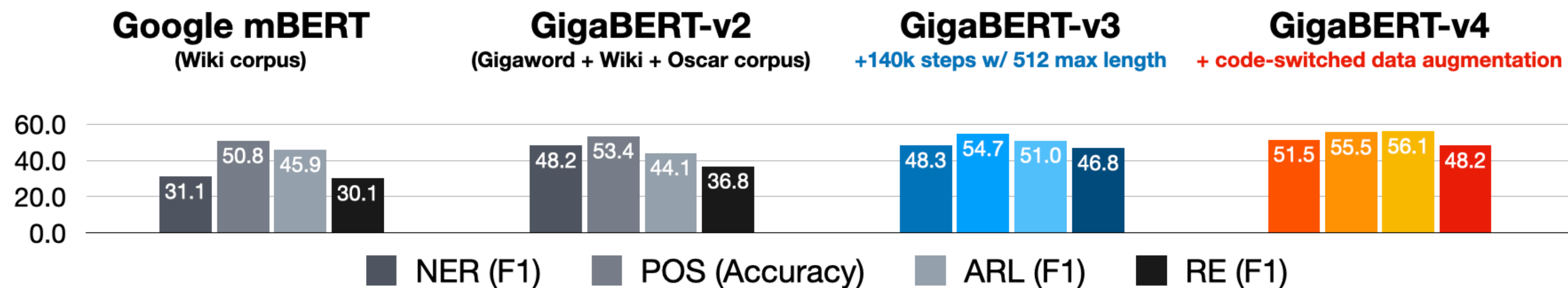


Bilingual BERT

- ▶ A customized bilingual BERT for Arabic NLP and English-to-Arabic zero-shot transfer learning + synthetic code-switching data

(English Translation — Official: China reaches the five-year plan target of cutting emissions)
مسؤول: الصين تبلغ هدف الخطة الخمسية المتعلق بخفض الانبعاثات

مسؤول: China تبلغ هدف plan الخمسية المتعلق بخفض emissions



mmBERT

- ▶ Multi-stage training with different number of languages per stage

Category	Dataset	Pre-training		Mid-training		Decay Phase	
		Tokens (B)	%	Tokens (B)	%	Tokens (B)	%
Code	Code (ProLong)	–	–	–	–	2.8	2.7
Code	Starcode	100.6	5.1	17.2	2.9	0.5	0.5
Crawl	DCLM	600.0	30.2	10.0	1.7	–	–
Crawl	DCLM (Dolmino)	–	–	40.0	6.7	2.0	2.0
Crawl	FineWeb2	1196.6	60.2	506.7	84.3	78.5	76.0
Instruction	Tulu Flan	15.3	0.8	3.1	0.5	1.0	1.0
Math	Dolmino Math	11.2	0.6	4.3	0.7	0.5	0.5
Reference	Books	4.3	0.2	3.9	0.7	2.2	2.1
Reference	Textbooks (ProLong)	–	–	–	–	3.1	3.0
Reference	Wikipedia (MegaWika)	4.7	0.2	1.2	0.2	9.5	9.2
Scientific	Arxiv	27.8	1.4	5.4	0.9	3.3	3.2
Scientific	PeS2o	8.4	0.4	3.2	0.5	–	–
Social	StackExchange	18.6	0.9	3.0	0.5	–	–
Social	StackExchange (Dolmino)	1.4	0.1	2.8	0.5	–	–
Total		1989.0	100.0	600.8	100.0	103.3	100.0

mmBERT

- Pre-trained on 3T tokens of multilingual text in over 1800 languages

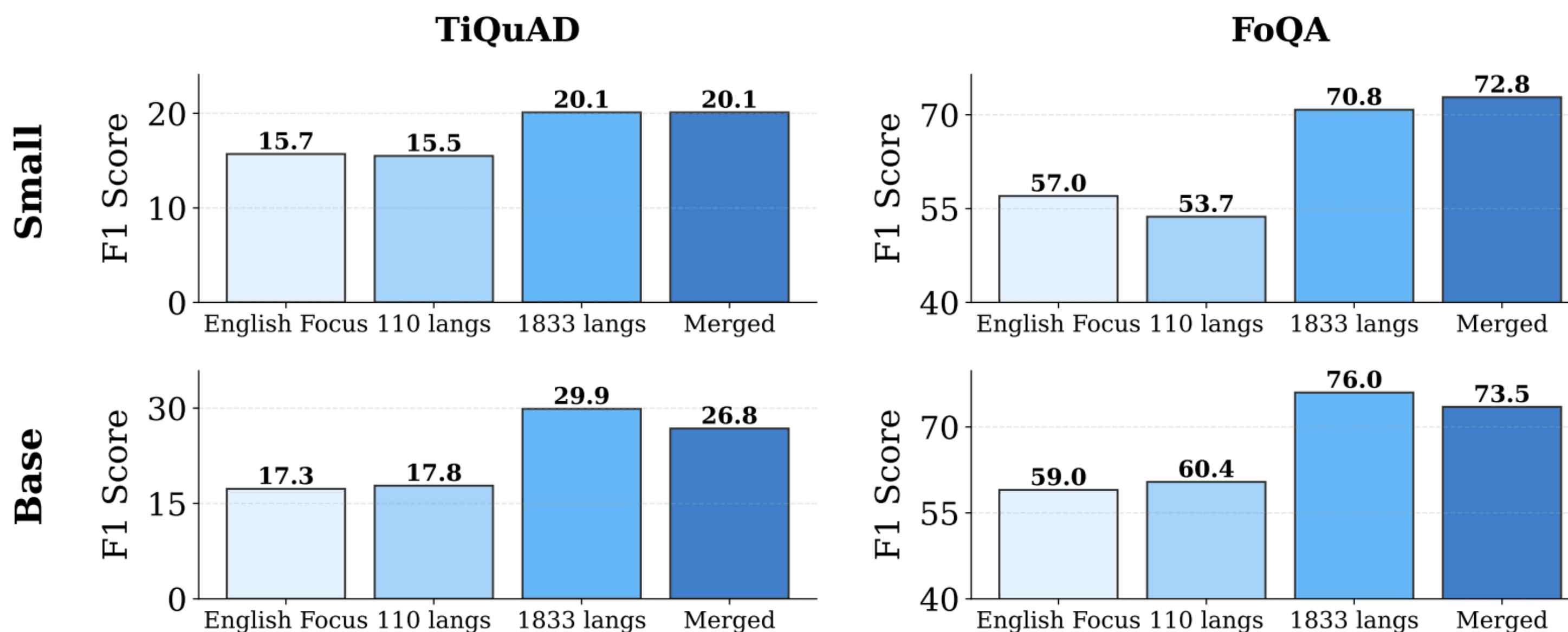


Figure 2: Performance of models using different decay phases on two languages (Tigray and Faroese) only added during the decay phase. We see that MMBERT with the 1833 language decay phase shows rapid performance improvements despite only having the models in the last 100B tokens of training. The final MMBERT models shows improvements by merging together checkpoints.

Other Encoder-only LMs

- ▶ XLM-RoBERTa (Conneau et al., 2020) for 100 languages
- ▶ mDeBERTa (He et al., 2021) for 100 languages
- ▶ MosaicBERT (Portes et al., 2023; Nussbaum et al., 2024)
- ▶ ModernBERT (Warner et al., 2024)
- ▶ EuroBERT (Boizard et al., 2025) for 15 languages
- ▶ NeoBERT (Le Breton et al., 2025)
- ▶ mmBERT (Marone et al., 2025) for 1800 languages
- ▶ Jev (TypeSafe AI, Sep 15 2026) ?

Compressing BERT

ALBERT

- ▶ A Lite BERT (18x fewer parameters, 1.7x faster training than BERT)
- ▶ Factorized embedding matrix to save parameters, model context-independent words with fewer parameters

Ordinarily $|V| \times H$ — $|V|$ is 30k-90k, H is >1000

-8	-2	-6	6
-4	-1	-3	3
28	7	21	-21
24	6	18	-18

 $=$

-2
-1
7
6

 $\times$

4	1	3	-3
---	---	---	----

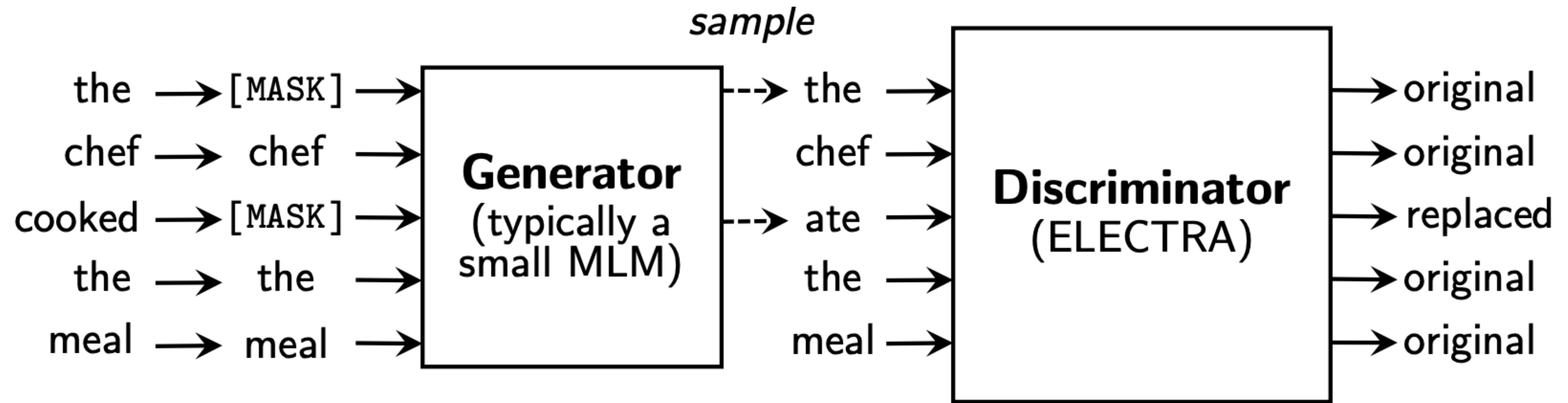
Factor into two matrices with a low-rank approximation

Now: $|V| \times E$ and $E \times H$ — E is 128 in their implementation

- ▶ Additional cross-layer parameter sharing

Lan et al. (2020)

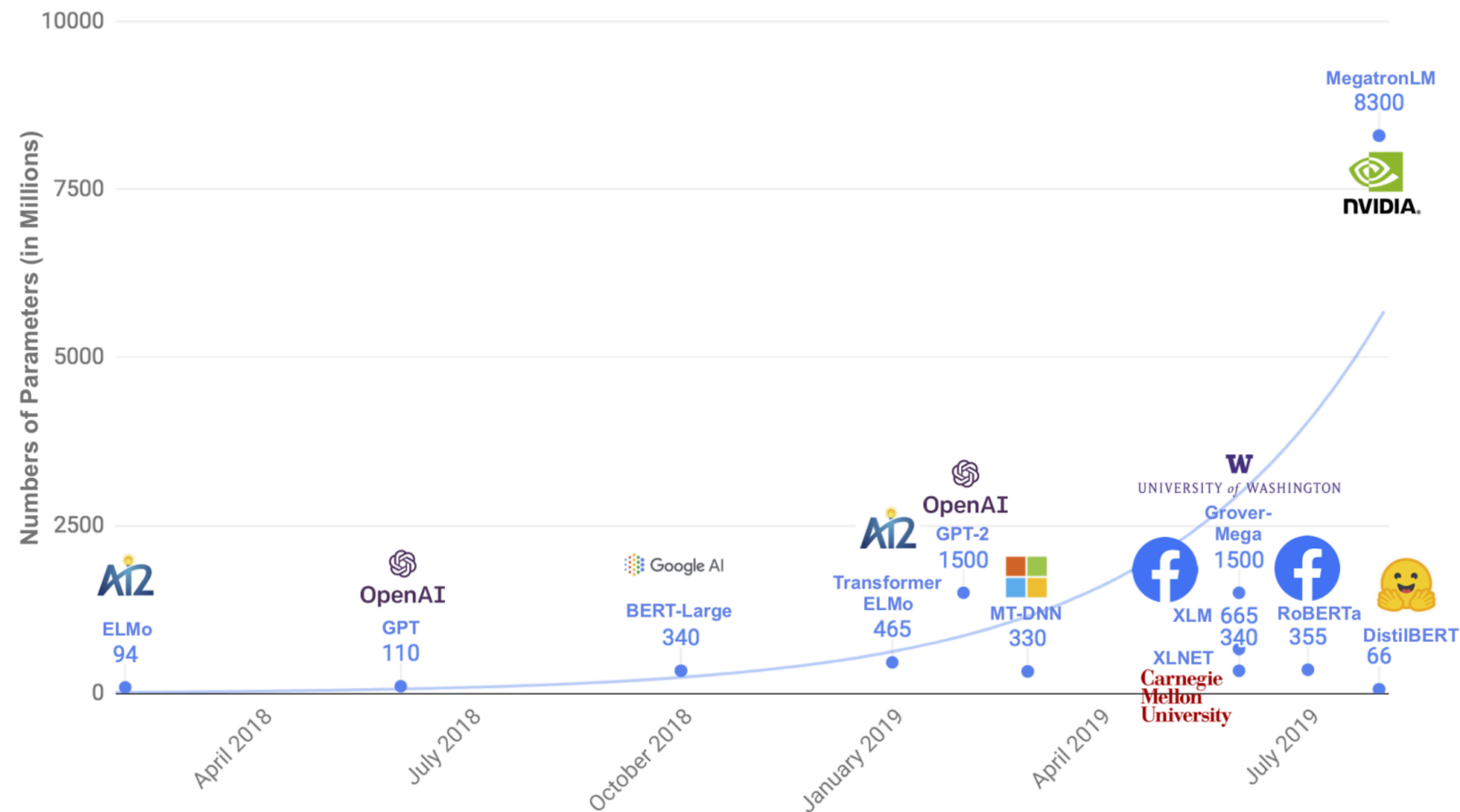
ELECTRA



- ▶ No need to necessarily have a generative model (predicting words)
- ▶ This objective is more computationally efficient (trains faster) than the standard BERT objective

DistilBERT

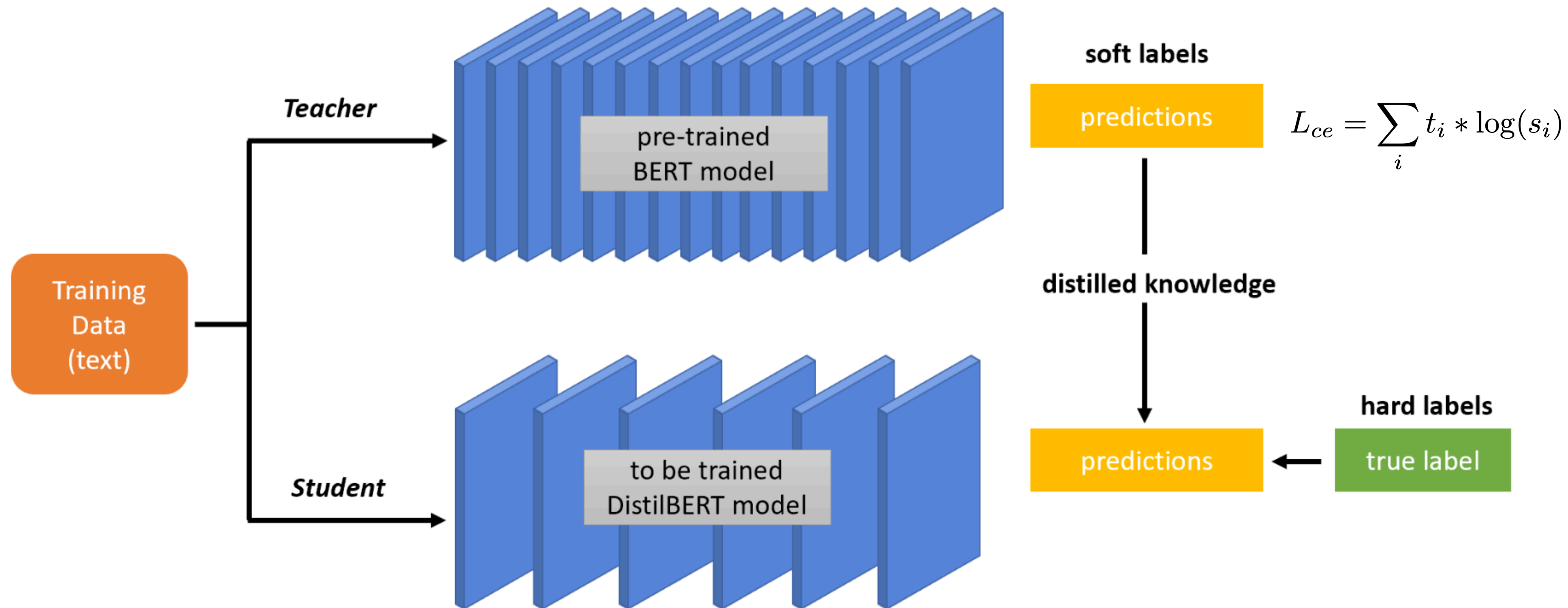
- ▶ Remove 60+% of BERT's heads post-training with minimal drop in performance
- ▶ DistilBERT (Sanh et al., 2019): nearly as good with half the parameters of BERT (via knowledge distillation)



Michel et al. (2019)

DistilBERT

- DistilBERT is pretrained by knowledge distillation to create a smaller model with faster inference and requires less compute to train.



Sanh et al. (2019)

Knowledge Distillation

Knowledge Distillation

arXiv:1503.02531v1 [stat.ML] 9 Mar 2015

Distilling the Knowledge in a Neural Network

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Abstract

A very simple way to improve the performance of almost any machine learning algorithm is to train many different models on the same data and then to average their predictions [3]. Unfortunately, making predictions using a whole ensemble of models is cumbersome and may be too computationally expensive to allow deployment to a large number of users, especially if the individual models are large neural nets. Caruana and his collaborators [1] have shown that it is possible to compress the knowledge in an ensemble into a single model which is much easier to deploy and we develop this approach further using a different compression technique. We achieve some surprising results on MNIST and we show that we can significantly improve the acoustic model of a heavily used commercial system by distilling the knowledge in an ensemble of models into a single model. We also introduce a new type of ensemble composed of one or more full models and many specialist models which learn to distinguish fine-grained classes that the full models confuse. Unlike a mixture of experts, these specialist models can be trained rapidly and in parallel.

1 Introduction

Many insects have a larval form that is optimized for extracting energy and nutrients from the environment and a completely different adult form that is optimized for the very different requirements of traveling and reproduction. In large-scale machine learning, we typically use very similar models for the training stage and the deployment stage despite their very different requirements: For tasks like speech and object recognition, training must extract structure from very large, highly redundant datasets but it does not need to operate in real time and it can use a huge amount of computation. Deployment to a large number of users, however, has much more stringent requirements on latency and computational resources. The analogy with insects suggests that we should be willing to train very cumbersome models if that makes it easier to extract structure from the data. The cumbersome model could be an ensemble of separately trained models or a single very large model trained with a very strong regularizer such as dropout [9]. Once the cumbersome model has been trained, we can then use a different kind of training, which we call “distillation” to transfer the knowledge from the cumbersome model to a small model that is more suitable for deployment. A version of this strategy has already been pioneered by Rich Caruana and his collaborators [1]. In their important paper they demonstrate convincingly that the knowledge acquired by a large ensemble of models can be transferred to a single small model.

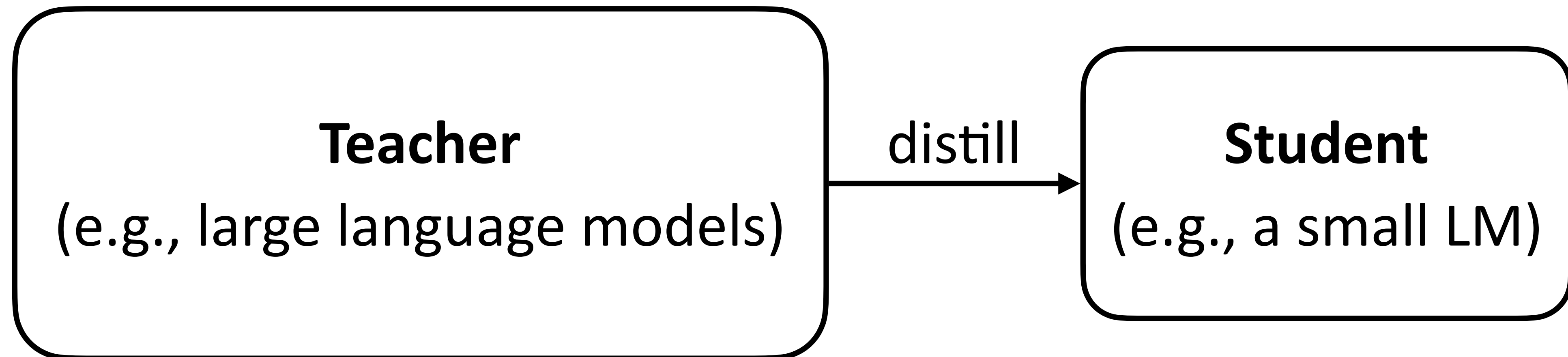
A conceptual block that may have prevented more investigation of this very promising approach is that we tend to identify the knowledge in a trained model with the learned parameter values and this makes it hard to see how we can change the form of the model but keep the same knowledge. A more abstract view of the knowledge, that frees it from any particular instantiation, is that it is a learned

^{*}Also affiliated with the University of Toronto and the Canadian Institute for Advanced Research.
[†]Equal contribution.

Hinton et al. (2015)

Knowledge Distillation

- ▶ Compress a large and complex model (the teacher model) into a smaller and simpler model (the student model)



(Recap) Multiclass Logistic Regression

$$P_w(y|x) = \frac{\exp(w^\top f(x, y))}{\sum_{y' \in \mathcal{Y}} \exp(w^\top f(x, y'))}$$

sum over output
space to normalize

i.e. minimize negative log likelihood
or **cross-entropy loss**

► Training: maximize $\mathcal{L}(x, y) = \sum_{j=1}^m \log P(y_j^* | x_j)$

index of
data points (j)

Knowledge Distillation

- Formally, NNs typically use a softmax function to convert logits z_i into class probabilities:

$$P(z_i, T) = \frac{\exp(z_i / T)}{\sum_j \exp(z_j / T)}$$

temperature to smooth softmax distribution

index of output space $\mathcal{Y}$

- One classic way of knowledge distillation is to align the class probability distribution from teacher and student networks.

Knowledge Distillation

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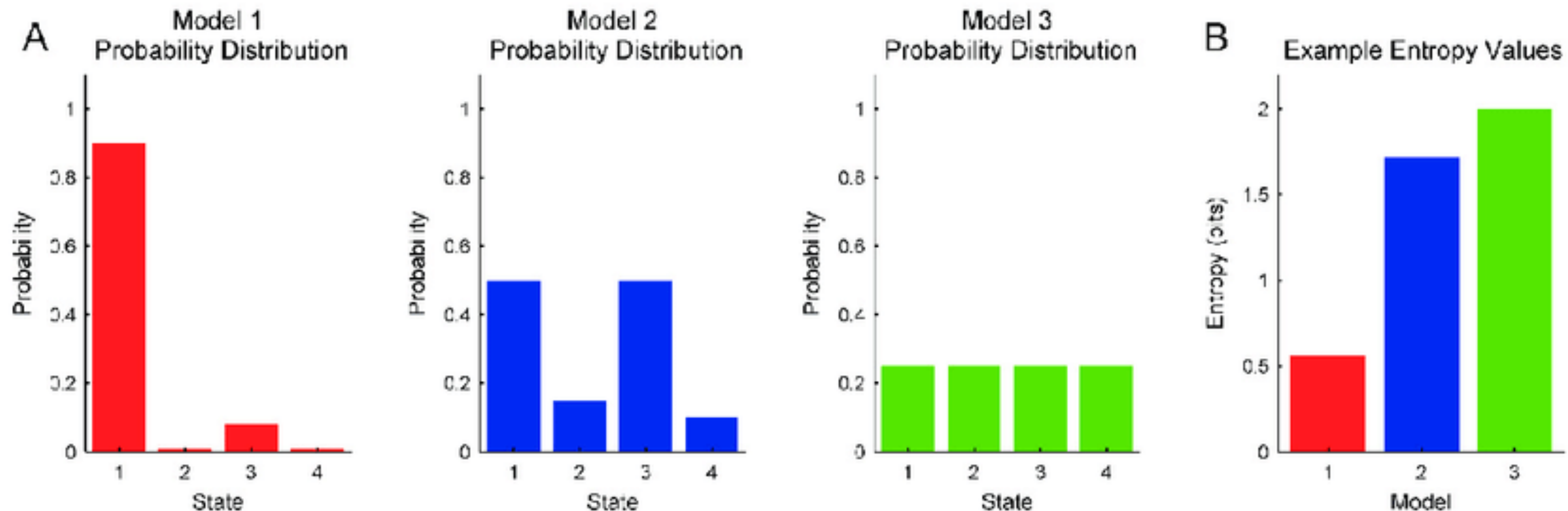
j index of output space $\mathcal{Y}$

- One classic way of knowledge distillation is to align the class probability distribution from teacher and student networks.
- More advanced KD include aligning intermediate weights, gradients of attention maps, sparsity patterns after ReLU activation, etc.

Entropy

- Entropy measures the inherent randomness or uncertainty of a single probability distribution

$$H(p) = - \sum_{i=1}^n p_i \log p_i$$



Cross-Entropy

- ▶ Cross Entropy (CE) is an extension of the concept of entropy, when two different probability distributions are present

$$H(p, q) = - \sum_{i=1}^n p_i \log q_i$$

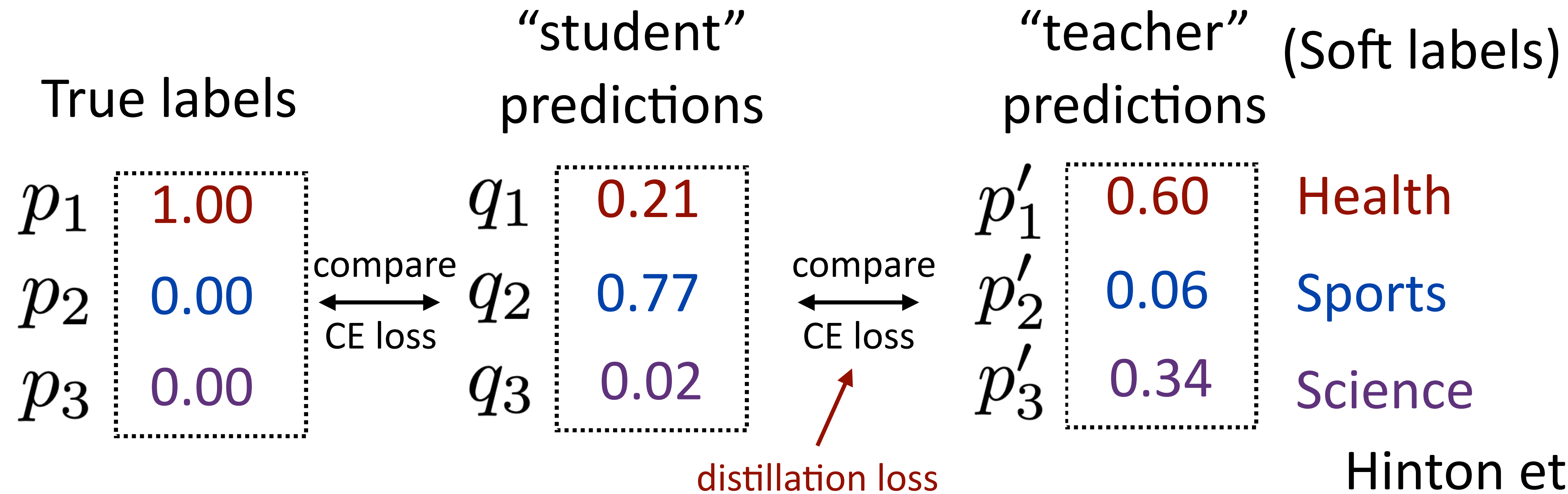
p_1	1.00
p_2	0.00
p_3	0.00

q_1	0.21
q_2	0.77
q_3	0.02

$$= - 1.00 * \log(0.21) = \log(0.21)$$

Distillation Loss

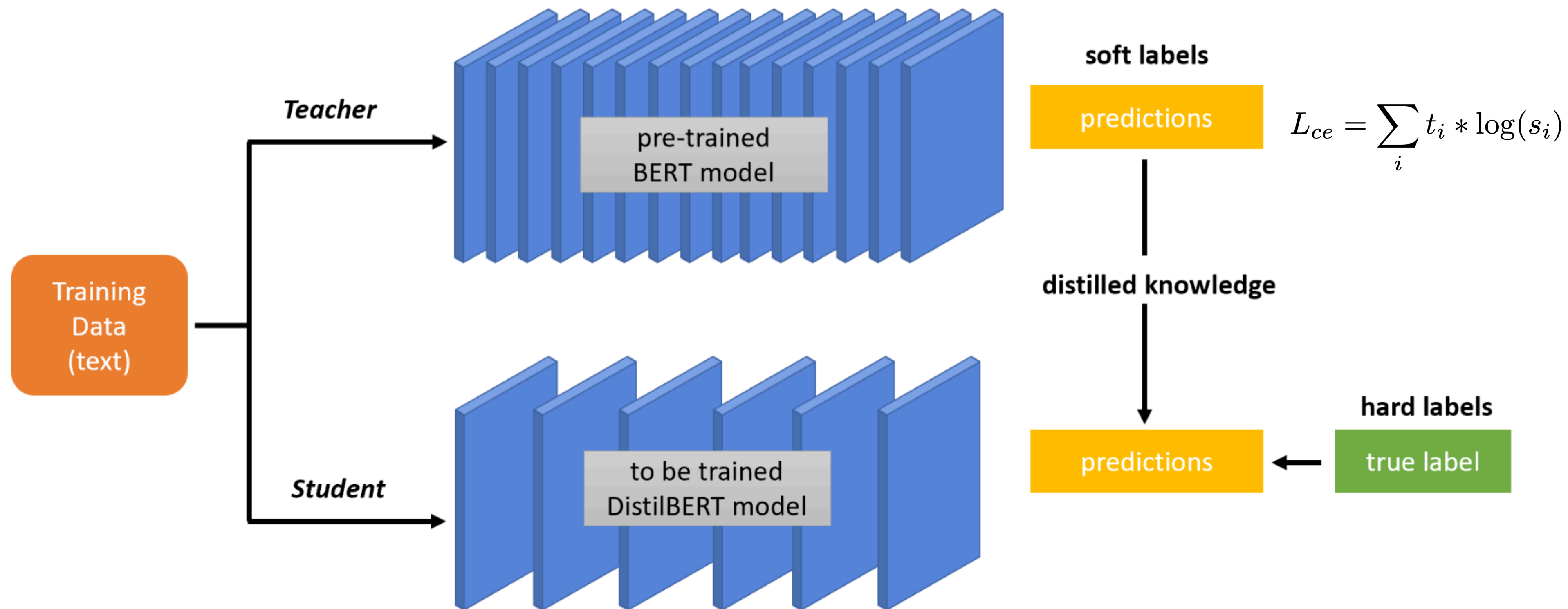
- ▶ Train student to mimic teacher's predicted distributions
- ▶ The relative probabilities of incorrect answers tell us a lot about how the large ("teacher") model tends to generalize (e.g., mixing up "Health/Science" is a smaller error than "Health/Sports").



Hinton et al. (2015)

DistilBERT

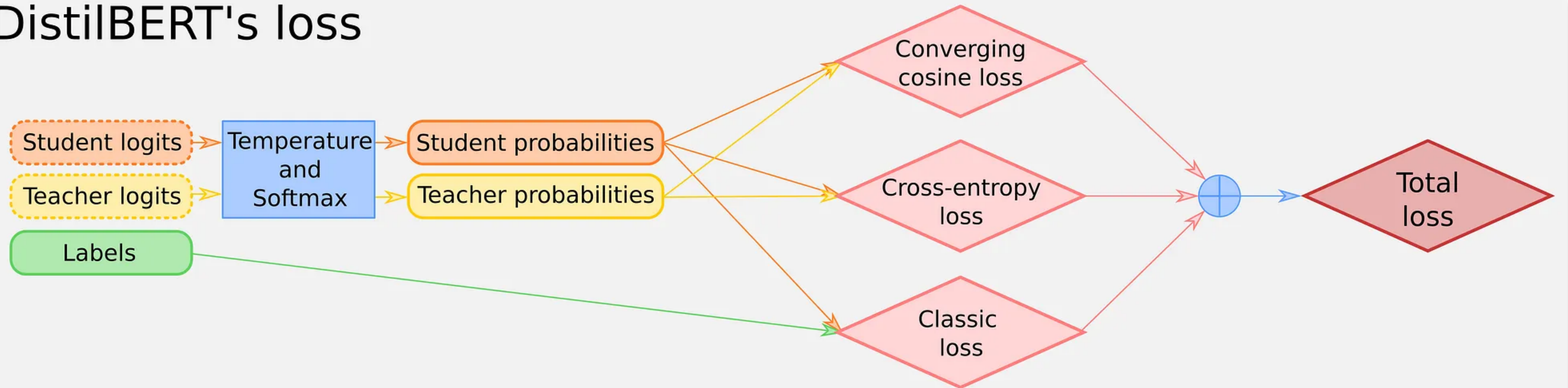
- DistilBERT is pretrained by knowledge distillation to create a smaller model with faster inference and requires less compute to train.



DistilBERT

- DistilBERT is pretrained by knowledge distillation to create a smaller model with faster inference and requires less compute to train.

DistilBERT's loss



Kullback–Leibler (KL) Divergence

- ▶ KL Divergence from p to q is defined as:

$$D_{\text{KL}}(p||q) = \sum_{i=1}^n p_i \log \frac{p_i}{q_i} = H(p, q) - H(p)$$

- ▶ If p is fixed (e.g., target distribution), minimizing cross-entropy $H(p, q)$ is equivalent to minimizing the KL divergence

Knowledge Distillation of LMs

- Token-level knowledge distillation

$$\mathcal{L}_{\text{token-KD}} = - \sum_{t=1}^L \sum_{v \in \mathcal{V}} \underset{\substack{\uparrow \\ \text{teacher}}}{p_T(v \mid \mathbf{y}_{<t}, \mathbf{x})} \log \underset{\substack{\uparrow \\ \text{student}}}{p_\theta(v \mid \mathbf{y}_{<t}, \mathbf{x})}$$

At each decoding step t , the student matches the teacher's next-token (or masked-token) prediction.

Knowledge Distillation of LMs

- Token-level knowledge distillation

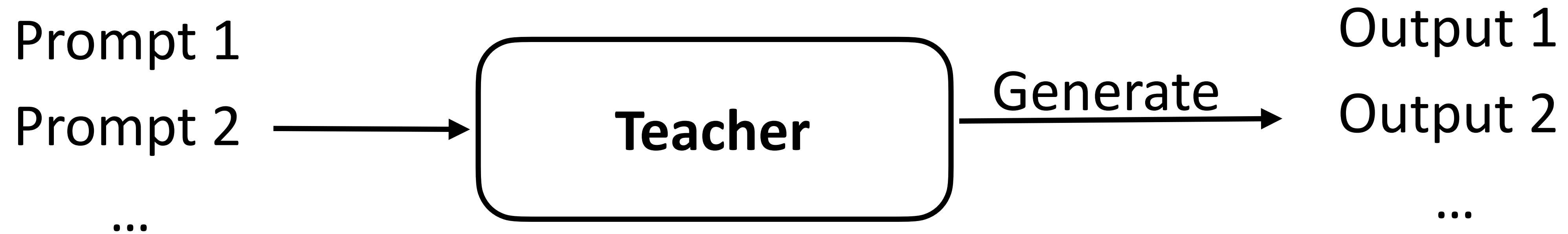
$$\mathcal{L}_{\text{token-KD}} = - \sum_{t=1}^L \sum_{v \in \mathcal{V}} \underset{\text{teacher}}{p_T(v \mid \mathbf{y}_{<t}, \mathbf{x})} \log \underset{\text{student}}{p_{\theta}(v \mid \mathbf{y}_{<t}, \mathbf{x})}$$

At each decoding step t , the student matches the teacher's next-token (or masked-token) prediction. This is equivalent to:

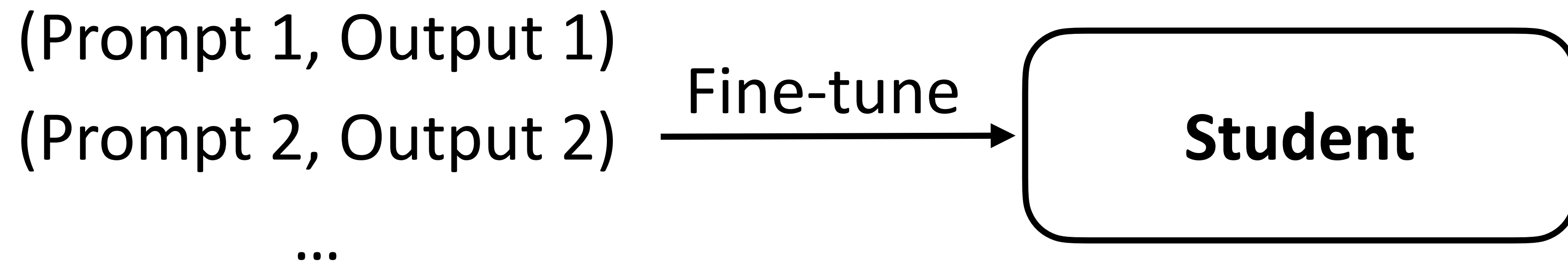
$$\mathcal{L}_{\text{token-KD}} = \sum_{t=1}^L \text{KL}(p_T(\cdot \mid \mathbf{y}_{<t}, \mathbf{x}) \parallel p_{\theta}(\cdot \mid \mathbf{y}_{<t}, \mathbf{x})) + \text{const.}$$

Sequence-level Knowledge Distillation

- ▶ Generate with a teacher model



- ▶ Student model fine-tunes on the data generated by the teacher model



Sequence-level Knowledge Distillation

- ▶ Generate with a teacher model

$$\hat{\mathbf{y}}_T = \arg \max_{\mathbf{y}} p_T(\mathbf{y} \mid \mathbf{x})$$

- ▶ Student model fine-tunes on the data generated by the teacher model

$$\mathcal{L}_{\text{seq-KD}} = -\log p_{\theta}(\hat{\mathbf{y}}_T \mid \mathbf{x}) = -\sum_{t=1}^{|\hat{\mathbf{y}}_T|} \log p_{\theta}(\hat{\mathbf{y}}_{T,t} \mid \hat{\mathbf{y}}_{T,<t}, x)$$

Treat $\hat{\mathbf{y}}_T$ as a pseudo-label and perform maximum likelihood training on it.
Kim & Rush (2016)

Knowledge Distillation of LMs

- ▶ Token-level knowledge distillation

$$p_T(\cdot \mid \mathbf{y}_{<t}, \mathbf{x}) \longrightarrow p_\theta(\cdot \mid \mathbf{y}_{<t}, \mathbf{x})$$

- ▶ Sequence-level knowledge distillation

$$\hat{\mathbf{y}}_T \sim p_T(\mathbf{y} \mid \mathbf{x}) \longrightarrow \max_{\theta} p_\theta(\hat{\mathbf{y}}_T \mid \mathbf{x})$$

Knowledge Distillation of LMs

- ▶ Token-level knowledge distillation

$$p_T(\cdot \mid \mathbf{y}_{<t}, \mathbf{x}) \longrightarrow p_\theta(\cdot \mid \mathbf{y}_{<t}, \mathbf{x})$$

$$\text{minimize } \text{KL}(p_T(\cdot \mid \mathbf{y}_{<t}, \mathbf{x}) \parallel p_\theta(\cdot \mid \mathbf{y}_{<t}, \mathbf{x}))$$

- ▶ Sequence-level knowledge distillation

$$\hat{\mathbf{y}}_T \sim p_T(\mathbf{y} \mid \mathbf{x}) \longrightarrow \max_{\theta} p_\theta(\hat{\mathbf{y}}_T \mid \mathbf{x})$$

minimize $\text{KL}(p_T(\mathbf{y} \mid \mathbf{x}) \parallel p_\theta(\mathbf{y} \mid \mathbf{x}))$ over all possible sequences — intractable

Teacher's distribution is approximated by one or more generated sequences

Takeaways

- ▶ Transformers + lots of data + self-supervision seems to do very well
- ▶ Knowledge distillation: Train a smaller model to imitate a stronger model
- ▶ Next time: GPT/GPT-2/GPT-3, etc.

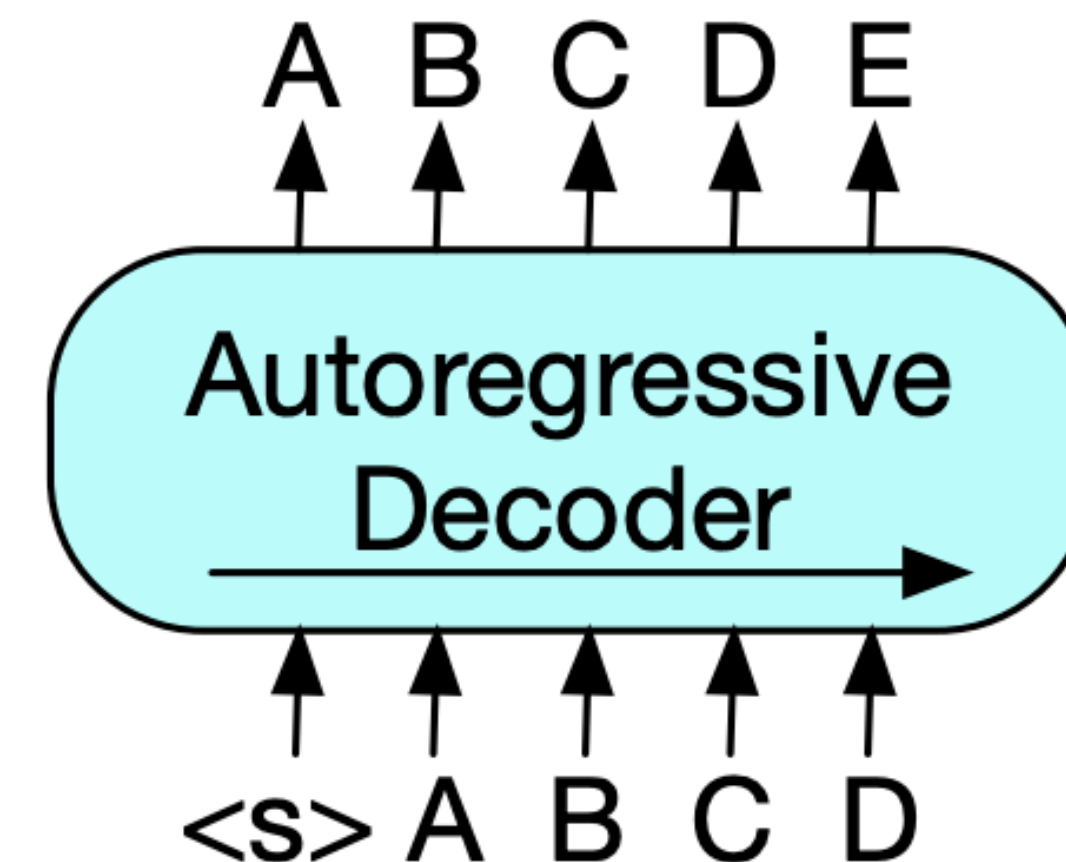
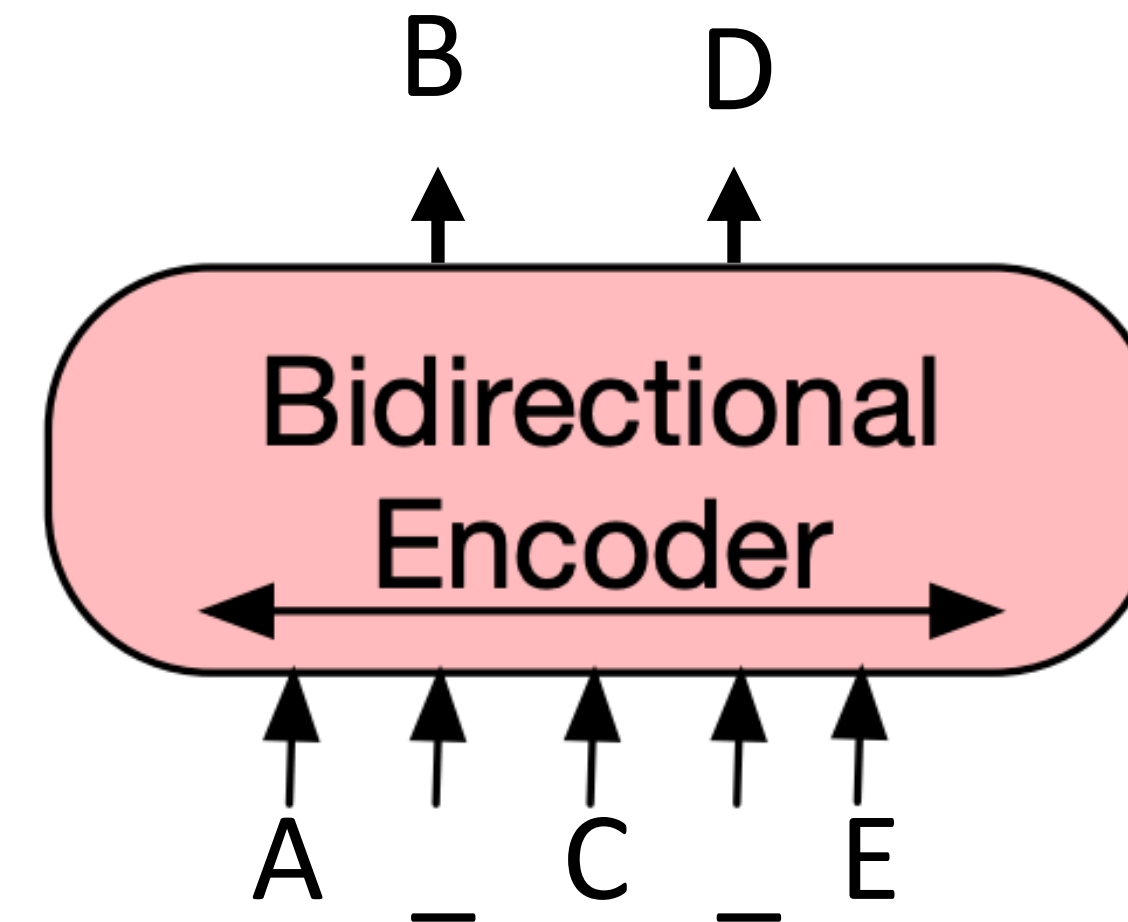
BERT/MLMs

- ▶ There are lots of ways to train these models!
- ▶ Key factors:
 - ▶ Big enough model
 - ▶ Big enough data
 - ▶ Well-designed “self-supervised” objective (something like language modeling). Needs to be a hard enough problem!

BART/T5
(encoder-decoder type of LMs)

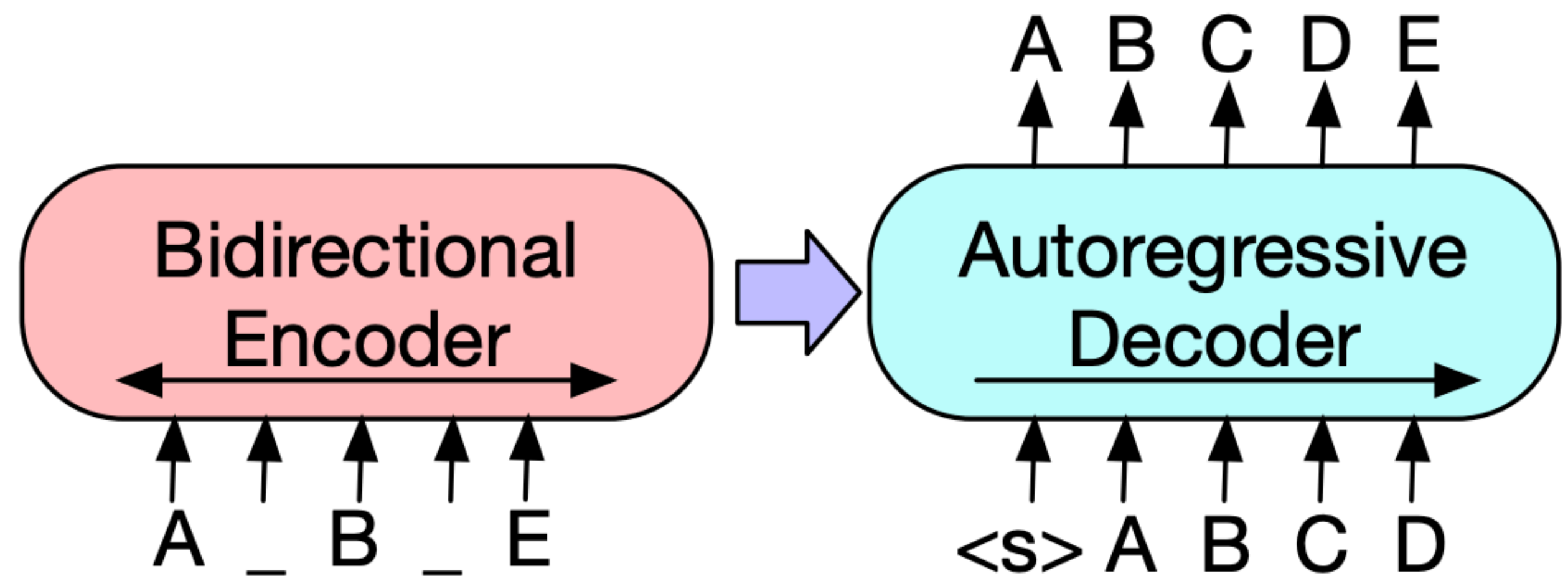
BERT (encoder) vs. GPT (decoder)

- ▶ BERT: only parameters are an encoder, trained with masked language modeling objective
 - ▶ No way to do translation or left-to-right language modeling tasks
- ▶ GPT: only the decoder, autoregressive LM
 - ▶ (Small-size versions) Typically used for unconditioned generation tasks, e.g. story or dialog generation



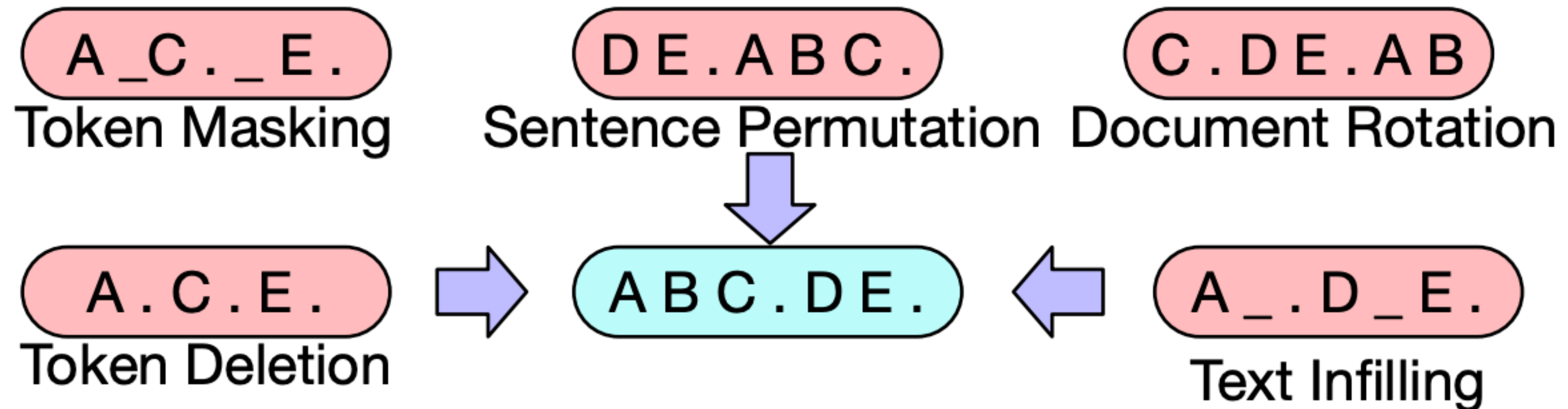
BART (encoder-decoder)

- ▶ What to do for seq2seq tasks?
- ▶ Sequence-to-sequence BERT variant: permute/make/delete tokens, then predict full sequence autoregressively
- ▶ For downstream tasks: feed document into both encoder + decoder, use decoder hidden state as output
- ▶ Good results on translation, summarization tasks



BART

- ▶ BART uses multiple de-noising LM objective:



Infilling is longer
spans than masking

BART

Model	SQuAD 1.1 F1	MNLI Acc	ELI5 PPL	XSum PPL	ConvAI2 PPL	CNN/DM PPL
BART Base						
w/ Token Masking	90.4	84.1	25.05	7.08	11.73	6.10
w/ Token Deletion	90.4	84.1	24.61	6.90	11.46	5.87
w/ Text Infilling	90.8	84.0	24.26	6.61	11.05	5.83
w/ Document Rotation	77.2	75.3	53.69	17.14	19.87	10.59
w/ Sentence Shuffling	85.4	81.5	41.87	10.93	16.67	7.89
w/ Text Infilling + Sentence Shuffling	90.8	83.8	24.17	6.62	11.12	5.41

- ▶ Infilling is all-around a bit better than masking or deletion
- ▶ Final system: combination of infilling and sentence permutation

Lewis et al. (2019)

SQuAD 2.0 (span-based QA)

- ▶ SQuAD 1.1 contains 100k+ QA pairs from 500+ Wikipedia articles.
- ▶ SQuAD 2.0 includes additional 50k questions that cannot be answered.
- ▶ These questions were crowdsourced.

Passage

Super Bowl 50 was an American football game to determine the champion of the National Football League (NFL) for the 2015 season. The **American Football Conference** (AFC) champion **Denver Broncos** defeated the National Football Conference (NFC) champion Carolina Panthers 24–10 to earn their third Super Bowl title. The game was played on February 7, **2016**, at Levi's Stadium in the San Francisco Bay Area at Santa Clara, California.

Question: Which NFL team won Super Bowl 50?

Answer: **Denver Broncos**

Question: What does AFC stand for?

Answer: **American Football Conference**

Question: What year was Super Bowl 50?

Answer: **2016**

BART

	SQuAD 1.1 EM/F1	SQuAD 2.0 EM/F1	MNLI m/mm	SST Acc	QQP Acc	QNLI Acc	STS-B Acc	RTE Acc	MRPC Acc	CoLA Mcc
BERT	84.1/90.9	79.0/81.8	86.6/-	93.2	91.3	92.3	90.0	70.4	88.0	60.6
UniLM	-/-	80.5/83.4	87.0/85.9	94.5	-	92.7	-	70.9	-	61.1
XLNet	89.0 /94.5	86.1/88.8	89.8/-	95.6	91.8	93.9	91.8	83.8	89.2	63.6
RoBERTa	88.9/ 94.6	86.5/89.4	90.2/90.2	96.4	92.2	94.7	92.4	86.6	90.9	68.0
BART	88.8/ 94.6	86.1/89.2	89.9/90.1	96.6	92.5	94.9	91.2	87.0	90.4	62.8

- Results on GLUE benchmark are not better than RoBERTa

BART

	SQuAD 1.1 EM/F1	SQuAD 2.0 EM/F1	MNLI m/mm	SST Acc	QQP Acc	QNLI Acc	STS-B Acc	RTE Acc	MRPC Acc	CoLA Mcc
BERT	84.1/90.9	79.0/81.8	86.6/-	93.2	91.3	92.3	90.0	70.4	88.0	60.6
UniLM	-/-	80.5/83.4	87.0/85.9	94.5	-	92.7	-	70.9	-	61.1
XLNet	89.0 /94.5	86.1/88.8	89.8/-	95.6	91.8	93.9	91.8	83.8	89.2	63.6
RoBERTa	88.9/ 94.6	86.5/89.4	90.2/90.2	96.4	92.2	94.7	92.4	86.6	90.9	68.0
BART	88.8/ 94.6	86.1/89.2	89.9/90.1	96.6	92.5	94.9	91.2	87.0	90.4	62.8

- Results on GLUE benchmark are not better than RoBERTa

CoLA

- ▶ Corpus of Linguistic Acceptability (CoLA); to test whether a model can recognize (a) morphological anomalies, (b) syntactic anomalies, and (c) semantic anomalies.

Label	Sentence	Source
*	The more books I ask to whom he will give, the more he reads.	Culicover and Jackendoff (1999)
✓	I said that my father, he was tight as a hoot-owl.	Ross (1967)
✓	The jeweller inscribed the ring with the name.	Levin (1993)
*	many evidence was provided.	Kim and Sells (2008)
✓	They can sing.	Kim and Sells (2008)
✓	The men would have been all working.	Baltin (1982)
*	Who do you think that will question Seamus first?	Carnie (2013)
*	Usually, any lion is majestic.	Dayal (1998)
✓	The gardener planted roses in the garden.	Miller (2002)
✓	I wrote Blair a letter, but I tore it up before I sent it.	Rappaport Hovav and Levin (2008)

(✓= acceptable, *=unacceptable)

Warstadt et al. (2020)

BART for Summarization

This is the first time anyone has been recorded to run a full marathon of 42.195 kilometers (approximately 26 miles) under this pursued landmark time. It was not, however, an officially sanctioned world record, as it was not an "open race" of the IAAF. His time was 1 hour 59 minutes 40.2 seconds. Kipchoge ran in Vienna, Austria. It was an event specifically designed to help Kipchoge break the two hour barrier.

Kenyan runner Eliud Kipchoge has run a marathon in less than two hours.

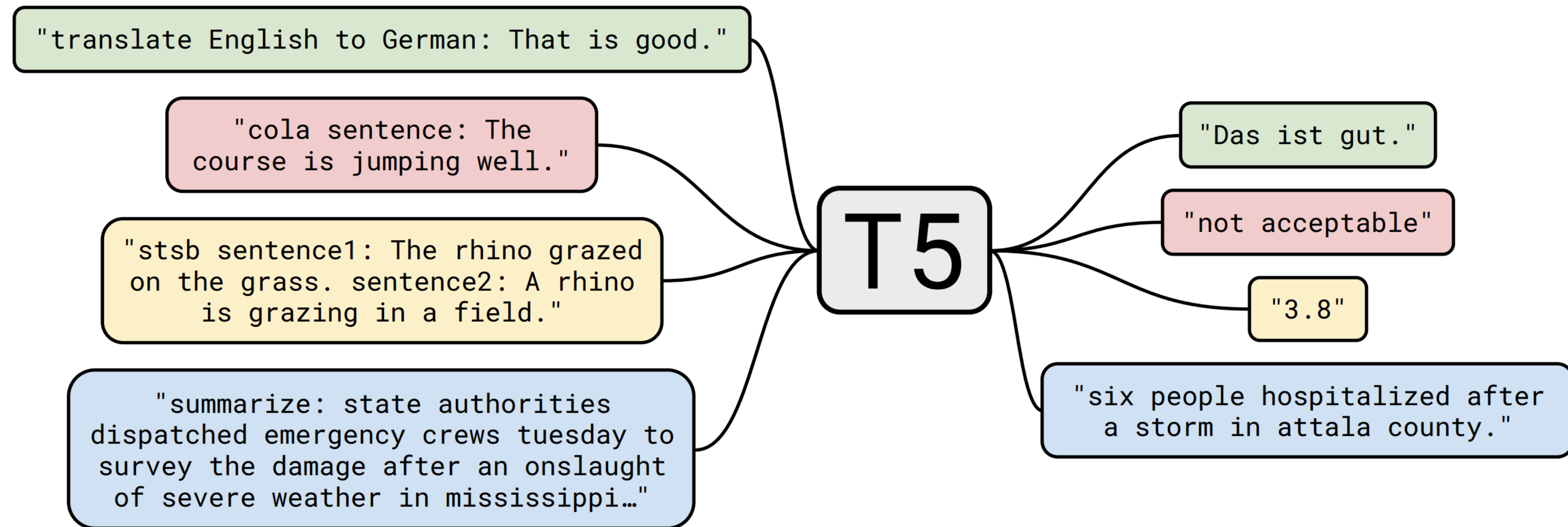
PG&E stated it scheduled the blackouts in response to forecasts for high winds amid dry conditions. The aim is to reduce the risk of wildfires. Nearly 800 thousand customers were scheduled to be affected by the shutoffs which were expected to last through at least midday tomorrow.

Power has been turned off to millions of customers in California as part of a power shutoff plan.

-
- But, strong results on dialogue, summarization, and other generation tasks.

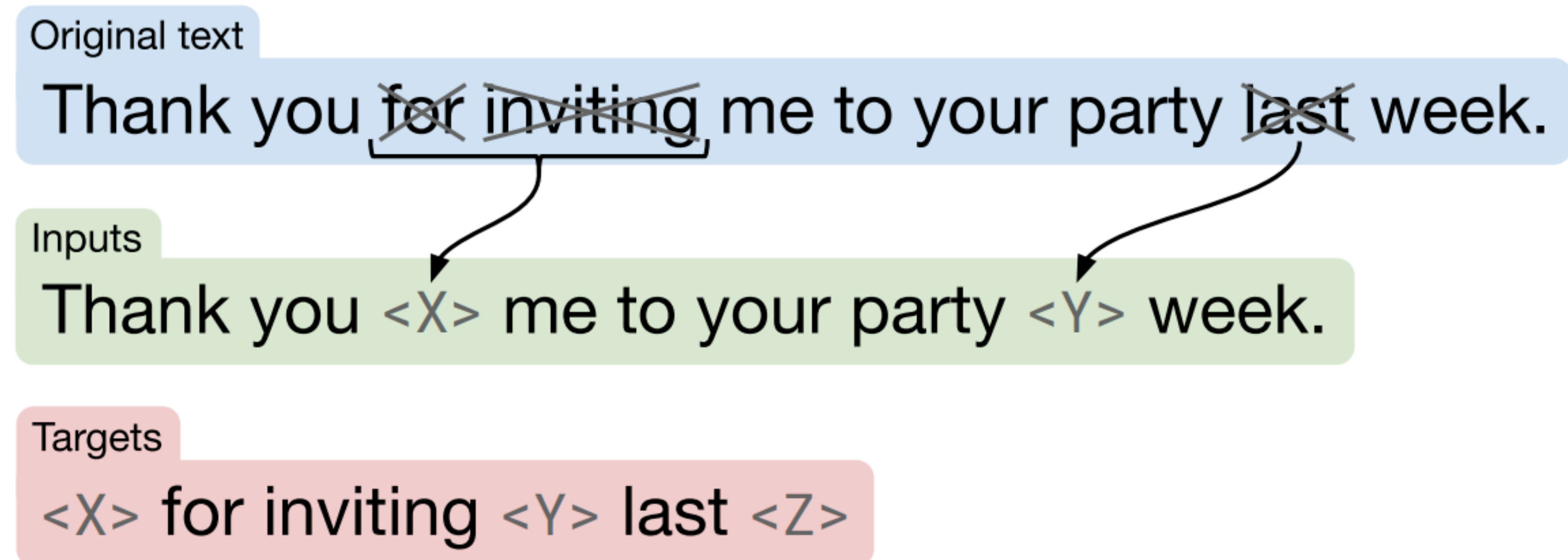
T5

- ▶ Frame many problems as sequence-to-sequence ones:



T5

- ▶ Pre-training: similar denoising scheme to BART



- ▶ Different mask tokens for individual masked spans; also different format for targets

T5

- Compared several different unsupervised LM objectives:

Objective	Inputs	Targets
Prefix language modeling	Thank you for inviting	me to your party last week .
BERT-style Devlin et al. (2018)	Thank you <M> <M> me to your party apple week .	<i>(original text)</i>
Deshuffling	party me for your to . last fun you inviting week Thank	<i>(original text)</i>
MASS-style Song et al. (2019)	Thank you <M> <M> me to your party <M> week .	<i>(original text)</i>
I.i.d. noise, replace spans	Thank you <X> me to your party <Y> week .	<X> for inviting <Y> last <Z>
I.i.d. noise, drop tokens	Thank you me to your party week .	for inviting last
Random spans	Thank you <X> to <Y> week .	<X> for inviting me <Y> your party last <Z>

T5

Number of tokens	Repeats	GLUE	CNNDM	SQuAD	SGLUE	EnDe	EnFr	EnRo
★ Full dataset	0	83.28	19.24	80.88	71.36	26.98	39.82	27.65
2^{29}	64	82.87	19.19	80.97	72.03	26.83	39.74	27.63
2^{27}	256	82.62	19.20	79.78	69.97	27.02	39.71	27.33
2^{25}	1,024	79.55	18.57	76.27	64.76	26.38	39.56	26.80
2^{23}	4,096	76.34	18.33	70.92	59.29	26.37	38.84	25.81

- ▶ Colossal Cleaned Common Crawl (C4): 750 GB of text
- ▶ We still haven't hit the limit of bigger data being useful for pre-training: here we see stronger MT results from the biggest data